

Element G

Construction of a Testable Prototype

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Neuqua Valley High School
PLTW Capstone

Problem Statement

Medical institutions lack a cost-effective, risk-free method for surgical training and other medical processes. Certified cadavers are the best training material for students, but are in great shortage worldwide, and pose ethical and practical issues. Due to said scarcity, organizations such as government agencies allocate a great proportion of their budget towards acquiring these materials: Medicare spends ~\$180,000 per medical resident each year through subsidies to universities, totaling over \$25 billion.¹ Overall, inefficient distribution and lack of product availability hinder surgical trainees worldwide, thus a more accessible solution is required.

I. Bill of Materials

The following tables describe the exact quantity and part type to build the Haptic VR Glove. It includes the name, description, cost, vendor, and additional notes. After drastically cutting down costs through 3D printing, the total cost for materials culminates to **\$146.95**.

Parts to Buy					
Item	Description	QTY	Cost	Vendor	Notes
Spandex Glove	Unisex Adult 18" Spandex Full Finger Gloves	1	\$9.99	Amazon	Pair
Servo Motors	MG90S servo with 360-degree rotation	1	\$20.99	Amazon	Pack
Micro Controller	Arduino Nano ESP32	1	\$22.00	Amazon	Board
Potentiometer	TWTADE 10Pcs Single Linear Rotary Seal Amplifier Potentiometers	1	\$12.66	Amazon	Pack
Retractable Badge Reel	JANYUN Black Retractable Badge Reels	1	\$12.99	Amazon	Pack
Batteries	Mini Portable Charger 5000mAh, USB-C Power Bank	3	\$7.99	Amazon	Singular
USB-C Ports	USB 3.1 Type C 24 Male and Female Plug Socket Adapter	1	\$7.49	Amazon	Pack
Wires	20 AWG Electric Wire Flexible Silicone Wires	1	\$14.99	Amazon	Box

¹ "The Costs of Training Internal Medicine Residents in the United States." The American Journal of Medicine, U.S. National Library of Medicine, pubmed.ncbi.nlm.nih.gov/25063649/. Accessed 12 Nov. 2024.

Velcro Strap	2 Rolls 32.2ft Straps 1 Inch Wide with 40 Metal Buckles	1	\$9.98	Amazon	Pack
Super Glue	48 x 3g Super Glue Clear	1	\$11.89	Amazon	Pack

Parts to Make			
Item	Description	QTY	Vendor
Screws	M1, M1.6, M2.5 Assorted Length Screws	1	NVHS & Ryan
Solder	Electrical Solder	1	NVHS
Elastic 3D Printer Filament	NinjaFlex	1	NVHS
Elastic Bands	Rubber Bands	1	NVHS
Oculus Quest 1	Meta Quest Generation 1 with Controllers	1	Ryan
3D Printed Part 1	Individual Servo Mount	5	NVHS
3D Printed Part 2	Potentiometer Tensioner	3	NVHS
3D Printed Part 3	Mirror Potentiometer Tensioner	2	NVHS
3D Printed Part 4	String Spool	5	NVHS
3D Printed Part 5	Spool Cover	5	NVHS
3D Printed Part 6	Micro Controller Holder	1	NVHS
3D Printed Part 7	Micro Controller Cover	1	NVHS
3D Printed Part 8	Top Battery Mount	1	NVHS
3D Printed Part 9	Bottom Battery Mount	1	NVHS
3D Printed Part 10	String Guide Cover	1	NVHS
3D Printed Part 11	Servo Base Plate	1	NVHS
3D Printed Part 12	Quest Controller Mount	1	NVHS
3D Printed Part 13	String Guide	14	NVHS
3D Printed Part 14	String Holster	5	NVHS
3D Printed Part 15	Thumb Top Mount	1	NVHS
3D Printed Part 16	Thumb Bottom Mount	1	NVHS
3D Printed Part 17	Pointer Top Mount	1	NVHS
3D Printed Part 18	Pointer Bottom Mount	1	NVHS
3D Printed Part 19	Middle Top Mount	1	NVHS
3D Printed Part 20	Middle Bottom Mount	1	NVHS
3D Printed Part 21	Ring Top Mount	1	NVHS

3D Printed Part 22	Ring Bottom Mount	1	NVHS
3D Printed Part 23	Pinky Top Mount	1	NVHS
3D Printed Part 24	Pinky Bottom Mount	1	NVHS
3D Printed Part 25	Servo Top Cover	1	NVHS

*all costs here are counted as \$0

Below are several diagrams of the Haptic VR Glove with all components shown and labeled.

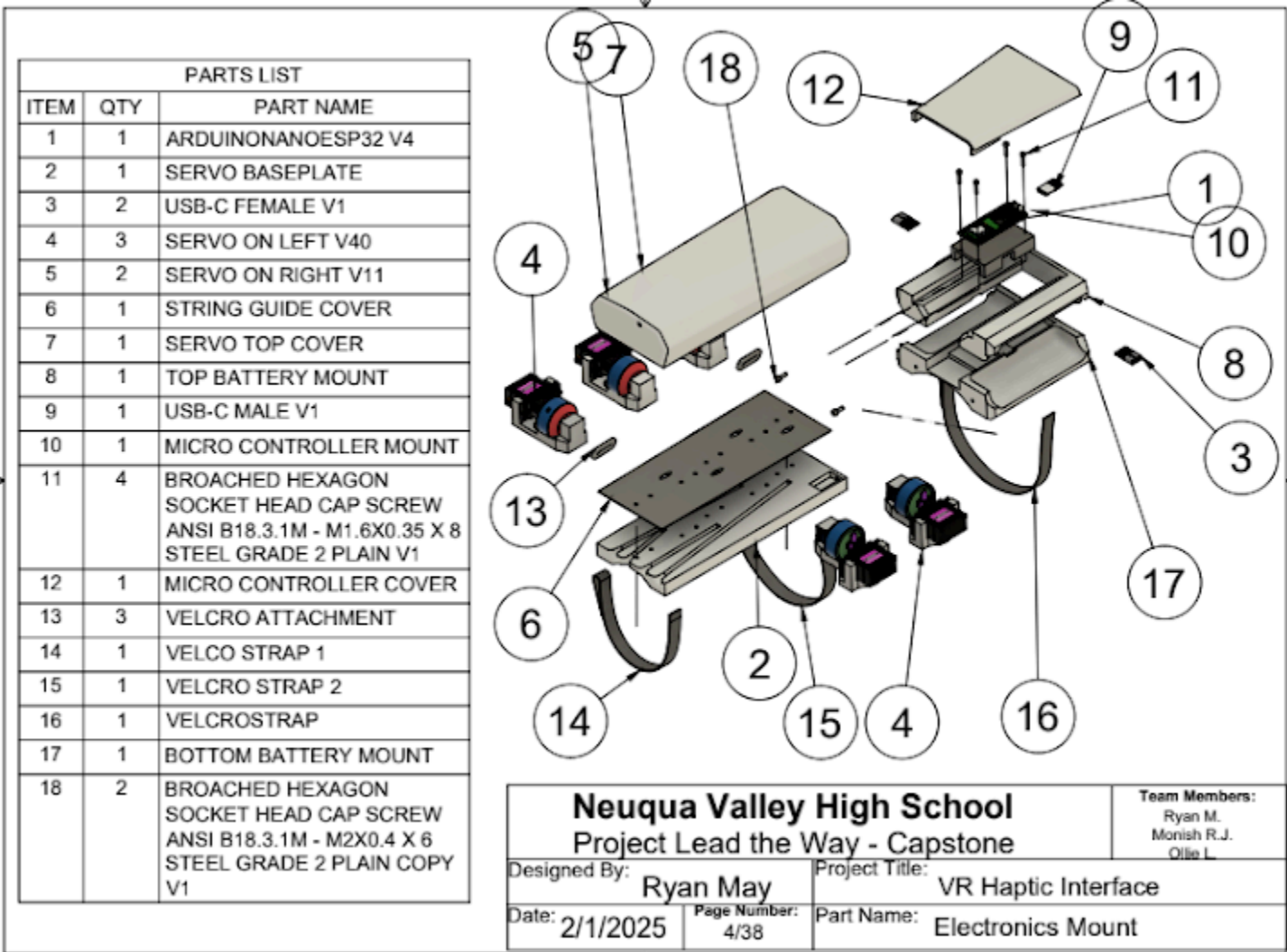


Figure 1.1: Exploded Electronics Mount labeled view

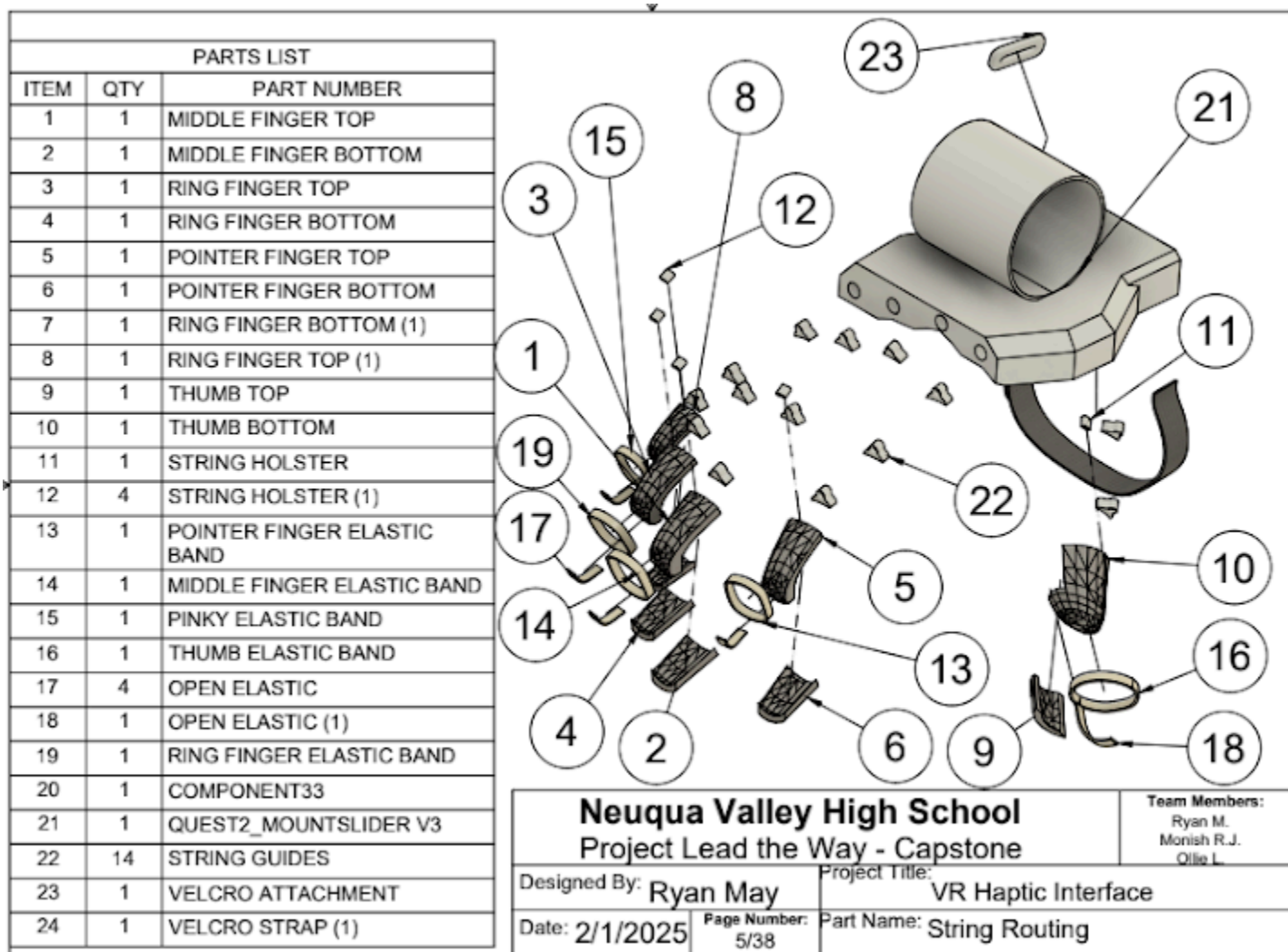
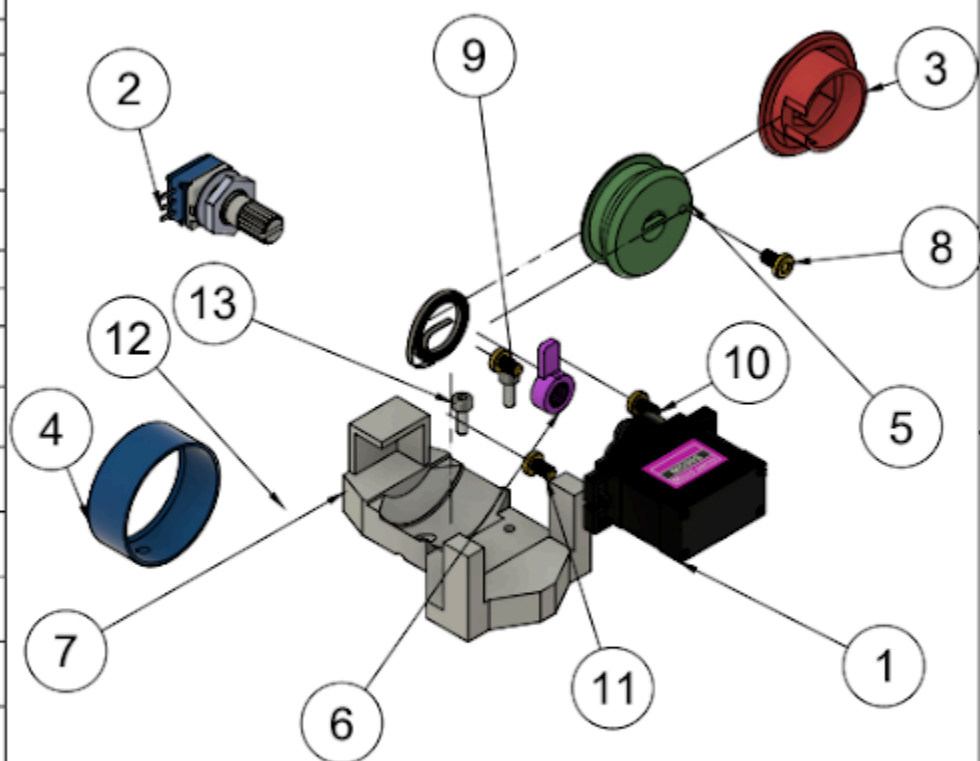


Figure 1.2: Exploded String Routing labeled view

PARTS LIST		
ITEM	QTY	PART NAME
1	1	ASSEMBLY SERVO V1
2	1	POTENTIOMETER
3	1	TENSIONER_GREENPO T V3
4	1	SPOOLCOVER_OPEN V2
5	1	HAPTICPOOL V3
6	1	SERVO ARM
7	1	INDIVIDUAL SERVO MOUNT
8	1	SCREW M2.5 X 6 CSK TORX BZP V1
9	1	SCREW M2.5 X 6 CSK TORX BZP V1 (1)
10	1	SCREW M2.5 X 6 CSK TORX BZP V1 (1) (1)
11	1	SCREW M2.5 X 6 CSK TORX BZP V1 (1) (1) (1)
12	1	SPOOLCOVER_OPEN V2 (1)
13	2	BROACHED HEXAGON SOCKET HEAD CAP SCREW ANSI B18.3.1M - M2X0.4 X 6 STEEL GRADE 2 PLAIN COPY V1



Neuqua Valley High School Project Lead the Way - Capstone		Team Members: Ryan M. Monish R.J. Ollie L.
Designed By:	Ryan May	Project Title:
Date:	2/1/2025	Part Name:
Page Number:	6/38	VR Haptic Interface
		Servo On Left

Figure 1.3: Exploded servo setup left side labeled view

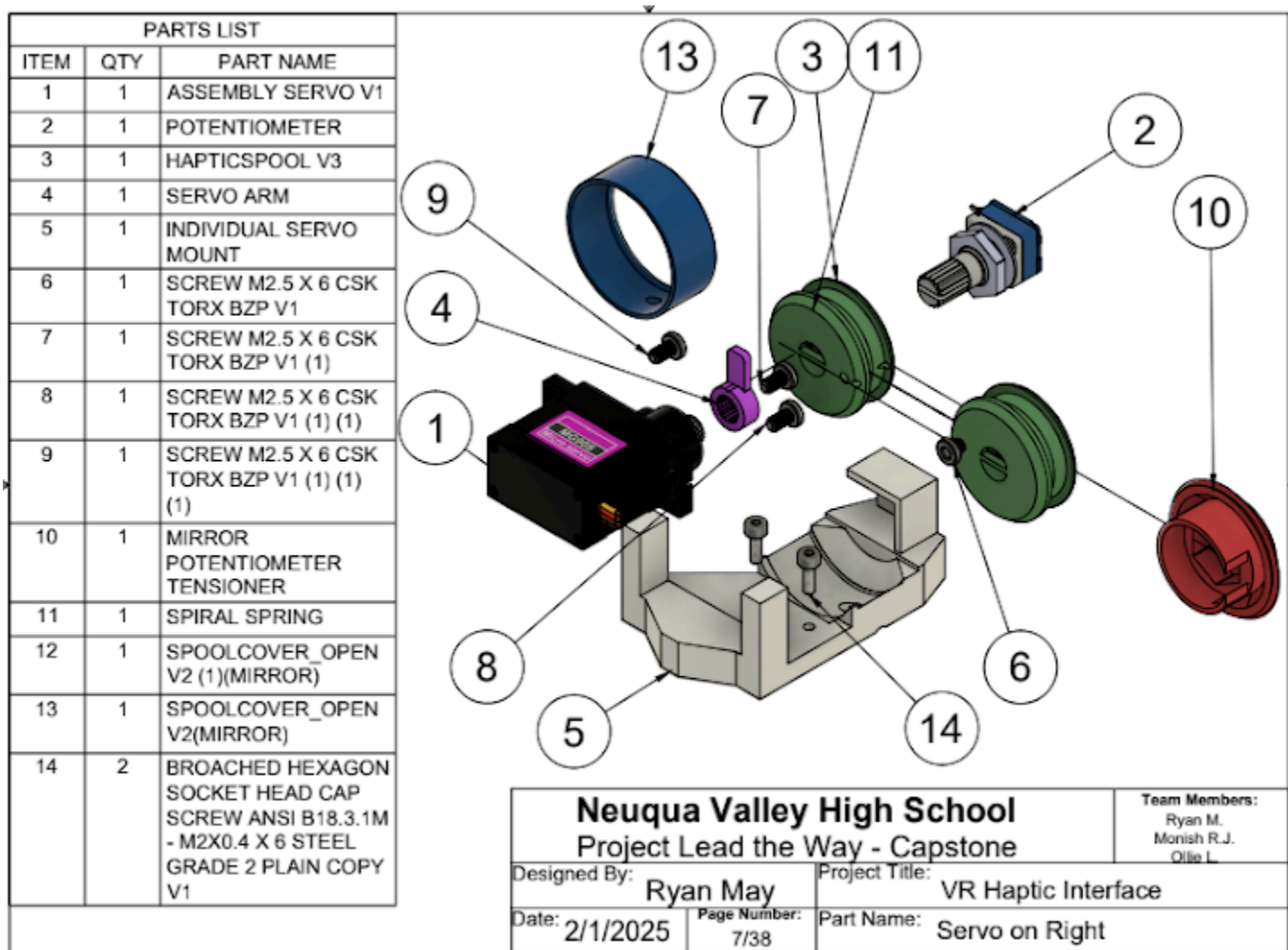


Figure 1.4: Exploded servo setup right side labeled view

II. Tools, Equipment, and Knowledge

Various tools and equipment are required to assemble the VR Haptic Glove, listed in the following table.

Tools	Location	Usage
Screwdriver	NVHS	Used for fastening screws
Soldering Iron	NVHS and Ryan	Used to solder wires to electronics and the microcontroller
Ruler	NVHS	Used to measure distances in metric and imperial
3D Printer	NVHS and Ryan	Used to print various base components and mounting structures
Wire Cutters	NVHS	Used to cut wires to the correct length for fingers
Flush Cutters	NVHS and Ryan	Used to trim 3D printed parts and open the ID Tag Spools
Super Glue/Tape	NVHS	Used to connect 3D printed parts and items to glove
Pen/Sharpie	NVHS	Used to mark positions on parts
Sandpaper	NVHS	Used to smooth out uneven 3D printed edges to ensure proper connection

Necessary knowledge from team members and potential outside knowledge are listed in the table below to assemble the VR Haptic Glove.

Ryan	Monish	Ollie	Outside Sources
- PLA 3D Printing - Soldering - Virtual Reality - Circuit Design - Plastic Welding	- Unity specialist - Main Programming - Construction - Simulation - Modeller	- Engineering principles - IMU knowledge - Sensor Integration - Basic programming - Assembly skills	- Advanced Programing - 3D Printing with Flexible Filament - Material Bonding - Testing

III. Build Procedure

Part Preparation

1. Using a 3D printer, print out the parts using PLA, or PETG, with appropriate slicer settings
2. Cut rubber bands so you end with 10 rubber band strips each one being 2 in. long. You will likely need to use multiple rubber bands.
3. Dismantle five badge reels for their spiral springs and thread.
4. Lubricate the Potentiometers
5. Using the Flush Cutters, carefully cut five of the retractable ID tag holders. You will need to insert the spiral spring and string inside each one.
6. Spray a small amount of WD-40 into the gap in the potentiometer and then rotate the potentiometer

7. 3D Print the following models in **PLA or PETG**:

- | | |
|--------------------------------------|------------------------------|
| a. 5x Individual Servo Mount | h. 1x Top Battery Mount |
| b. 3x Potentiometer Tensioner | i. 1x Bottom Battery Mount |
| c. 2x Mirror Potentiometer Tensioner | j. 1x Servo Top Cover |
| d. 5x String Spool | k. 1x String Guide Cover |
| e. 5x Spool Cover | l. 1x Servo Base Plate |
| f. 1x Micro Controller Holder | m. 1x Quest Controller Mount |
| g. 1x Micro Controller Cover | n. 14x String Guide |
| | o. 5x String Holster |

3D Print the following models in **NinjaFlex**:

- | | |
|-------------------------|------------------------|
| a. Thumb Top Mount | f. Middle Bottom Mount |
| b. Thumb Bottom Mount | g. Ring Top Mount |
| c. Pointer Top Mount | h. Ring Bottom Mount |
| d. Pointer Bottom Mount | i. Pinky Top Mount |
| e. Middle Top Mount | j. Pinky Bottom Mount |

Servo on Left

1. Place the hex nut (included with the potentiometer set) into the inside cutout of the Potentiometer Tensioner.

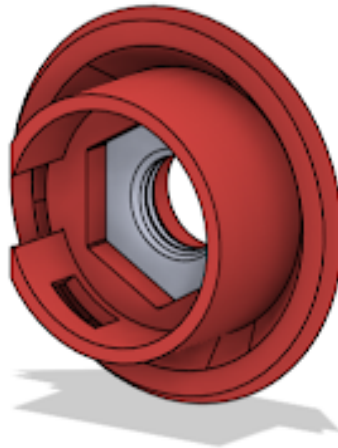


Figure 3.1: Hex nut (Gray) placed inside the Potentiometer Tensioner (Red).

2. Place a washer onto the shelf of the potentiometer, then screw the Potentiometer Tensioner, with the nut, into the threads on the potentiometer.

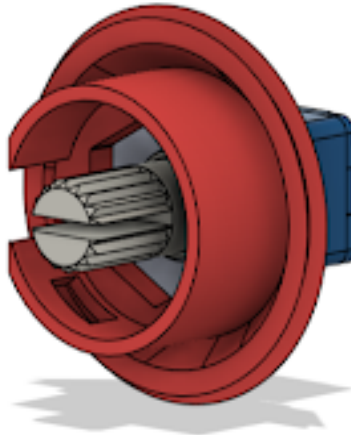


Figure 3.2: Potentiometer Tensioner (Red) connected to the potentiometer (Gray and Blue).

3. Align the small cutout of the badge reel spring into the small hole (Figure 3.3.a) in the cylinder of the potentiometer tensioner. Then, rotate the spring 90 degrees to connect both (Figure 3.3.b).

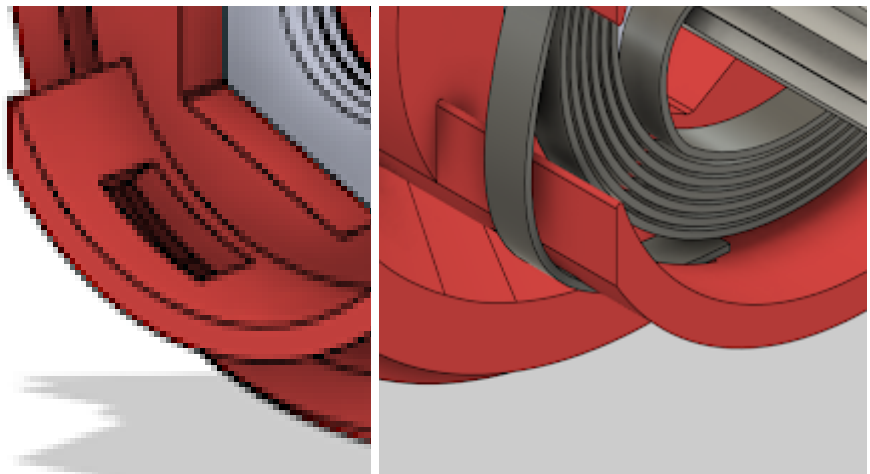


Figure 3.3.a (left): Spring hole in Potentiometer Tensioner (Red).

3.4.b (right): Spiral Spring (Dark Gray) in Potentiometer Tensioner (Red).

4. Place the rest of the coiled spring inside the potentiometer tensioner's cylinder.

5. Thread the other end of the spring through the slit in the potentiometer's shaft, connecting both. See Figure 3.4 for the final view.

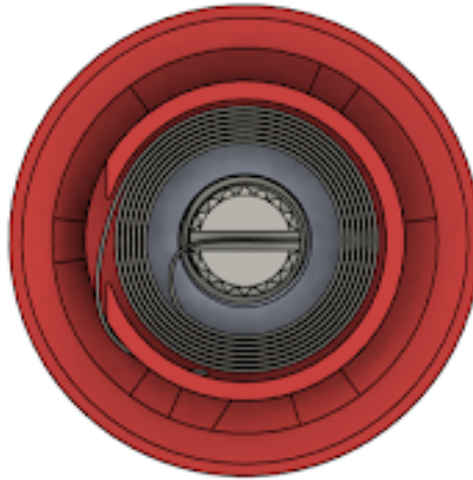


Figure 3.4: Spiral Spring (Dark Gray) completely encased in Potentiometer Tensioner (Red). When rotated, the potentiometer should spring to its neutral position on release.

6. Thread the loose end of the badge reel string (from Preparation) through the hole in the bottom of the Individual Servo Mount.

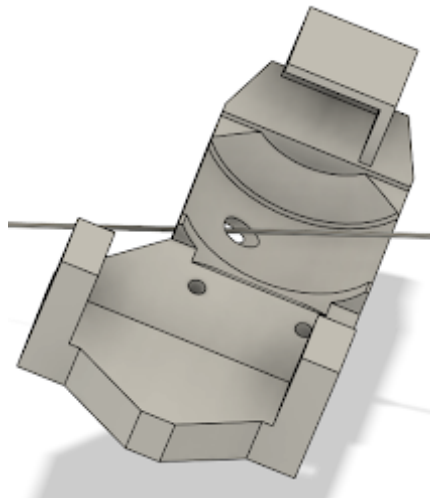


Figure 3.5: Servo Mount (Gray) with the string threaded through it.

7. Thread the same loose end of the string through the spool cover.

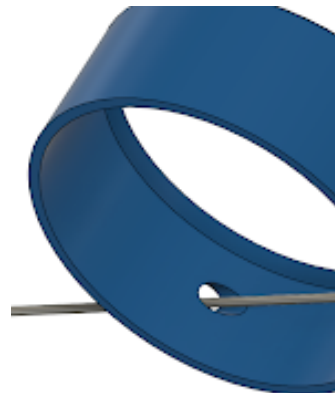


Figure 3.6: Spool Cover (Blue) with the loose end of the string threaded through it.

8. Thread the same loose end of the string through the hole in the String Spool.

9. Tie a knot on the loose end of the string on the outside of the Spool so the spool and string are now connected (Figure 3.7).

10. Wind the String around the spool until ~12 in. of the string remains unwinded (Figure 3.8).

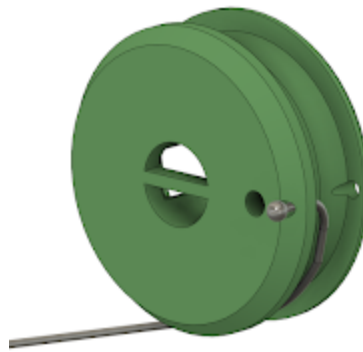


Figure 3.7: Connected string with String Spool (Green) loose end of string threaded through it and knotted.

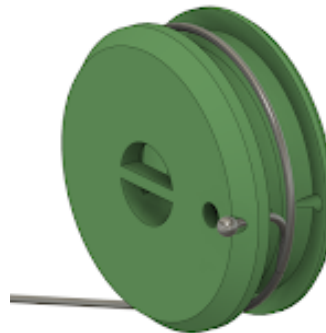


Figure 3.8: Loaded String Spool (Green).

11. Place the corresponding opening in the String Spool into the Potentiometer, then push it in until it is against the Spring Tensioner (Figure 3.10.b).

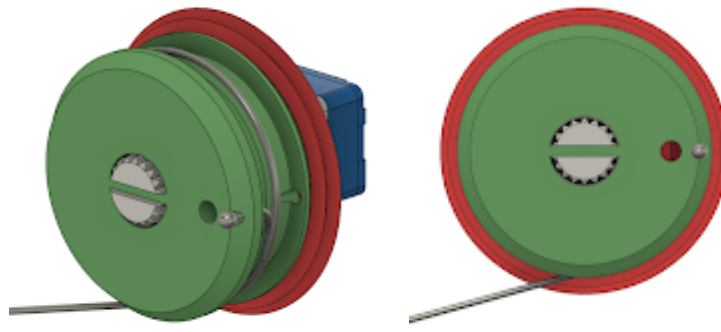


Figure 3.9.a (left): Connected String Spool (Green) and Potentiometer Tensioner (Red).

Figure 3.9.b (right): Top view of all connected components.

12. Move the spool cover into place around the String Spool.

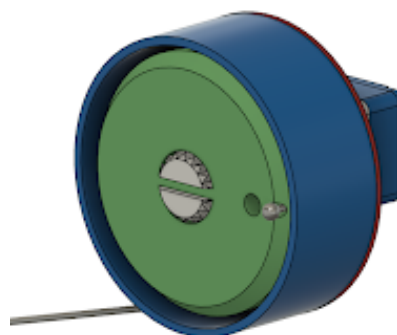


Figure 3.10: Spool Cover (Blue) surrounding string mechanism.

13. Insert all of these parts into the Individual Servo Mount so that the Potentiometer is snugly seated in it.

14. Take an M2.5 screw and screw it far enough into the second hole in the Spool Mount such that only 5mm of the screw is sticking out of the spool.

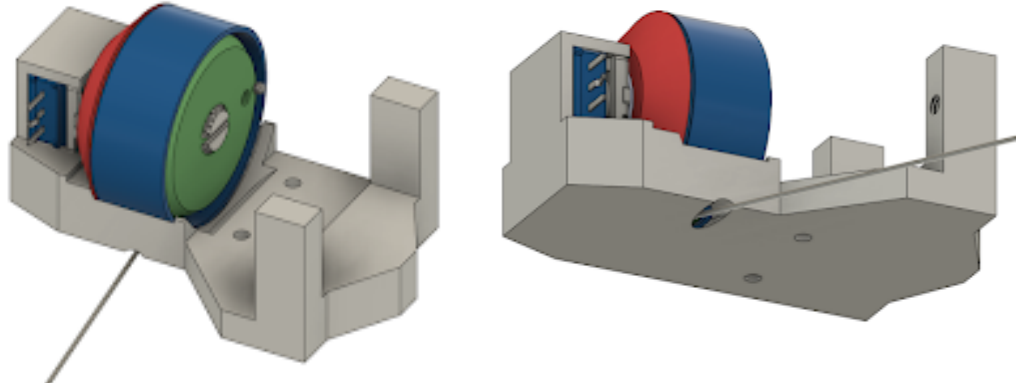


Figure 3.11.a (left): Fitted components (Red/Green/Blue) in Servo Mount (Gray).

Figure 3.11.b (right): Bottom isometric view.

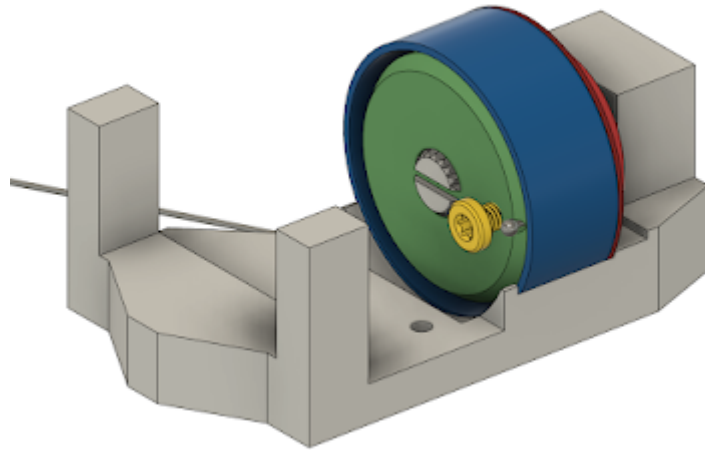


Figure 3.12: Potentiometer restrictor as M2.5 Screw (Yellow).

15. Place the servo arm onto the servo so that the rotation of the servo causes the arm to move, then screw them together with the screw that came with the servo.

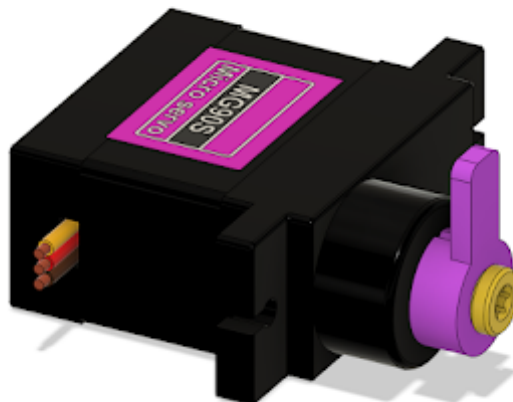


Figure 3.13: Attached servo arm (Purple) onto the base motor (Black) with the screw (Yellow).

16. Move the servo and servo arm such that it is between the two standing prongs.

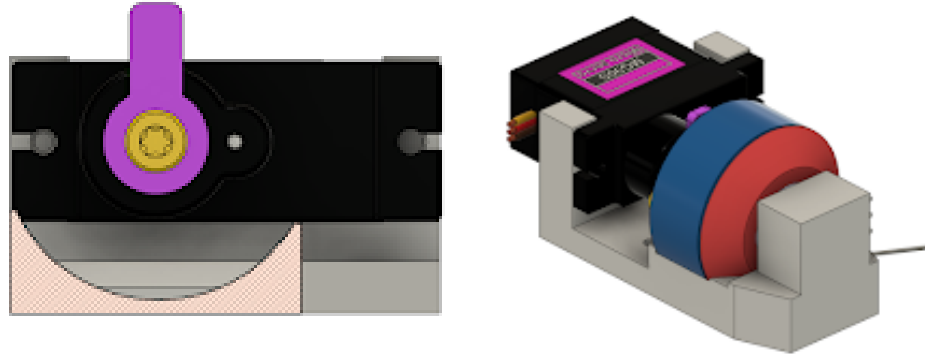


Figure 3.14.a (left): Servo arm (Purple) height relative to the motor base (Black).

Figure 3.14.b (right): Servo (Black) positioned correctly in servo mount (Gray).

17. Use a M2 x 6mm screw and fasten the servo to the Individual servo mount.

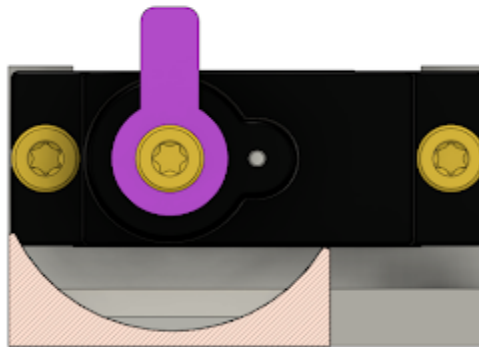


Figure 3.15: Connected servo (Black) onto servo mount (Gray) using screws (Yellow).

Final Assembly Appearance

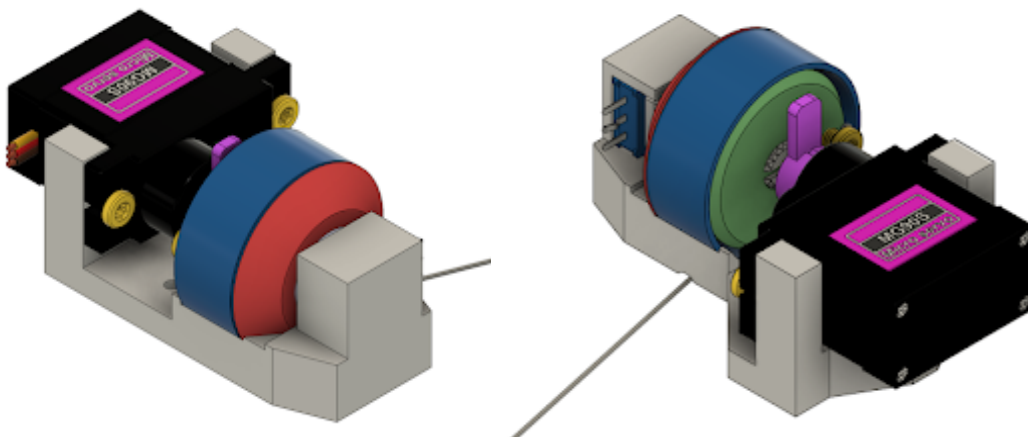


Figure 3.16.a (left): Isometric view of completed Servo on Left.

Figure 3.16.b (right): Alternate isometric view of completed Servo on Left.

The component Servo on Left is now completed. Repeat steps 1-17 twice until 3 total Servo on Left assemblies have been constructed.

Servo on Right

Info before starting:

- Servo on Right is almost exactly like Servo on Left, except mirrored
 - The string is coming out the other direction.
 - The component uses the mirrored Potentiometer Tensioner and Individual Servo Mount.

1. Place the Hex Nut that comes with the potentiometer into the cutout on the inside of the Mirror Potentiometer Tensioner.

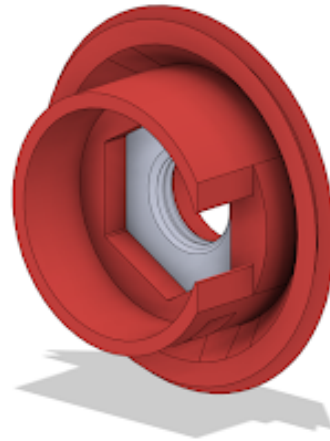


Figure 4.1: Hex nut (Gray) placed inside the Potentiometer Tensioner (Red).

2. Place the washer that comes with the potentiometer on the shaft of the potentiometer.

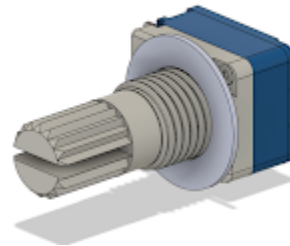


Figure 4.2: Washer (Gray) encircling potentiometer screw (Blue).

3. With the Nut, screw both the Nut and Mirror Potentiometer Tensioner into the threads on the potentiometer, until it can not easily rotate any farther.

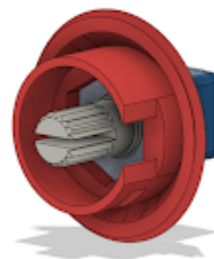


Figure 4.3: Potentiometer Tensioner (Red) connected to the potentiometer (Gray and Blue).

4. Insert the side of the spring with the small cutout in the sides of it into the small hole in the cylinder on the mirror potentiometer tensioner, then rotate it 90 degrees so they are connected.

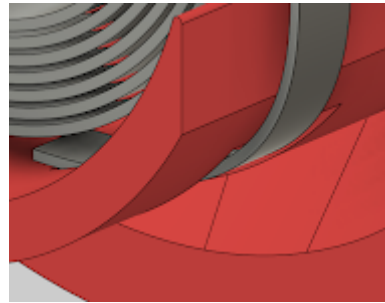


Figure 4.4: Spring hole in Potentiometer Tensioner (Red).

5. Place the rest of the coiled spring inside of the mirror potentiometer tensioner's cylinder.

6. Put the other end of the spring through the slit in the potentiometer's shaft so it is connected.

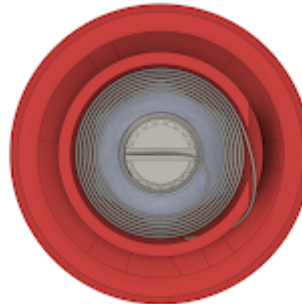


Figure 4.5: Spiral Spring (Dark Gray) completely encased in Potentiometer Tensioner (Red). When rotated, the potentiometer should spring to its neutral position on release.

7. Thread the loose end of the string through the hole in the bottom of the Individual Servo Mount, and bring the Individual Servo Mount to the other side of the string.

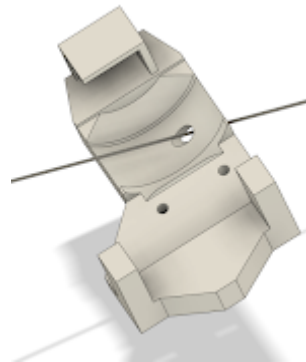


Figure 4.6: Servo Mount (Gray) with the string threaded through it.

8. Thread the loose end of the string through the spool cover, and bring the spool cover to the other side of the string.



Figure 4.7: Spool Cover (Blue) with the loose end of the string threaded through it.

9. Thread the loose end of the string through the hole in the String Spool.

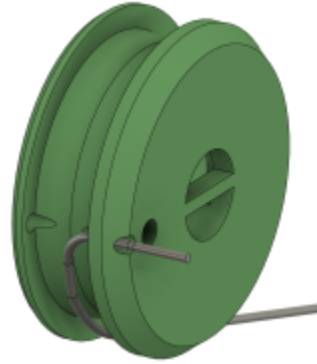


Figure 4.8: Connected string with String Spool (Green) loose end of string threaded through it and knotted.

10. Tie a knot on the loose end of the string on the outside of the Spool so the spool and string are now connected.

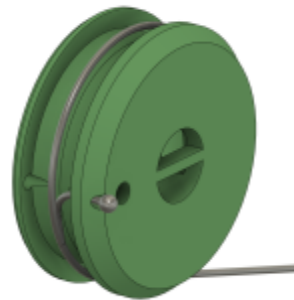


Figure 4.9: Loaded String Spool (Green).

12. Place the corresponding opening in the String Spool into the slit in the Potentiometer, then push it in until it is against the Mirror Spring Tensioner.

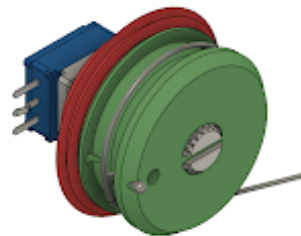


Figure 4.10: Connected String Spool (Green) and Potentiometer Tensioner (Red).

13. Move the Spool Cover around the String Spool.

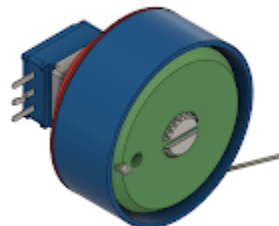


Figure 4.11: Spool Cover (Blue) surrounding string mechanism.

14. Insert all of these parts into the Individual Servo Mount such that the Potentiometer is seated snugly into the mount.

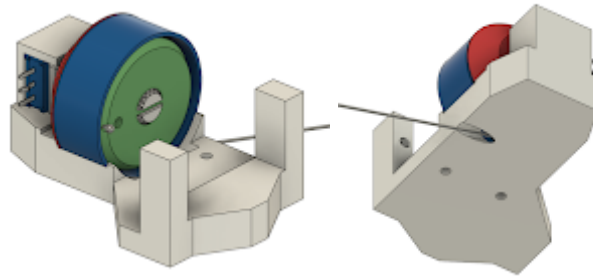


Figure 4.12.a (left): Fitted components (Red/Green/Blue) in Servo Mount (Gray).

Figure 4.12.b (right): Bottom isometric view.

15. Take an M2.5 screw and screw it far enough into the second hole in the Spool Mount such that only 5mm of the screw is sticking out of the spool.

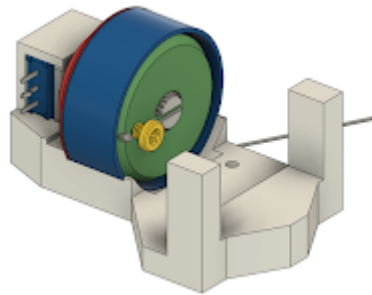


Figure 4.13: Potentiometer restrictor as M2.5 Screw (Yellow).

16. Place the Servo arm onto the servo so that the rotation of the servo causes the arm to move, then screw them together with the screw that came with the servo.

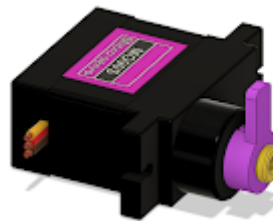


Figure 4.14: Attached servo arm (Purple) onto the base motor (Black) with the screw (Yellow).

17. Insert the servo and servo arm such that it is between the two standing prongs.



Figure 4.15: Servo arm (Purple) height relative to the motor base (Black).

18. Use a M2 x 6mm screw and fasten the servo to the Individual Servo Mount.

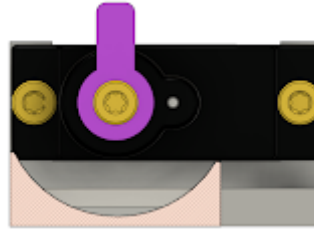


Figure 4.16: Connected servo (Black) onto servo mount (Gray) using screws (Yellow).

Final Assembly Appearance

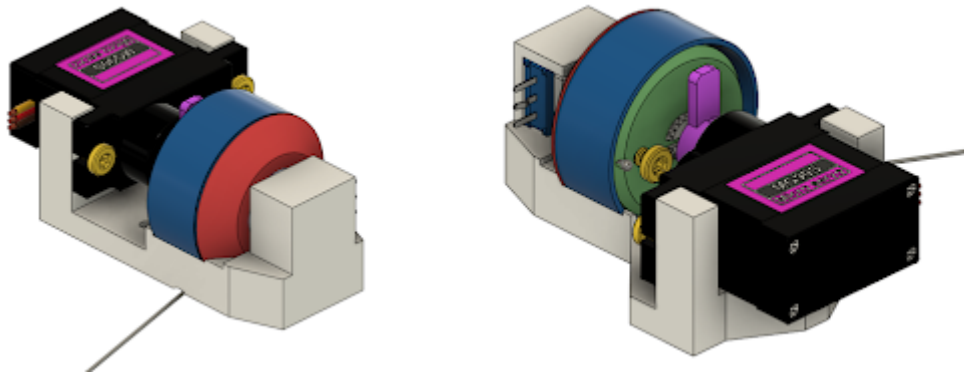


Figure 4.17.a (left): Isometric view of completed Servo on Right.

Figure 4.17.b (right): Alternate isometric view of completed Servo on Right.

The component Servo on Right is now completed. Repeat steps 1-17 until 2 total Servo on Right assemblies have been constructed.

Electronics Holder

1. Cut the end off of each string's exposed end.

2. Take one Servo on Right, thread the loose end of the string through the string hole on the longest parallel size of the String Guide Cover, and pull it through.

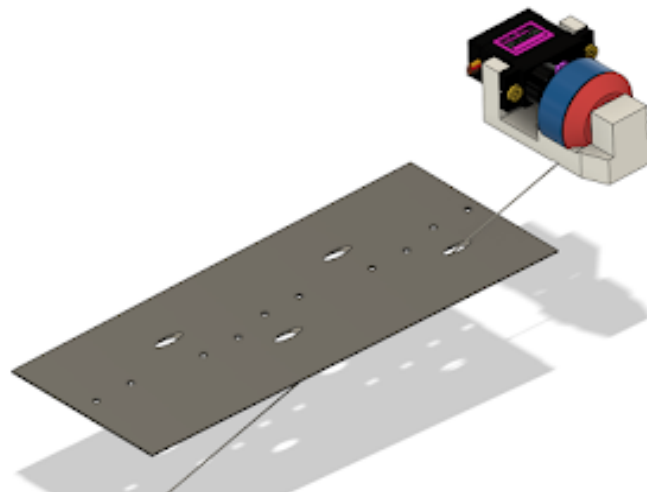


Figure 5.1: Servo on Right string threaded through the String Guide Cover.

3. Place the Servo on Right on the String Guide Cover, lining the corresponding screw holes on the String Guide Cover together.

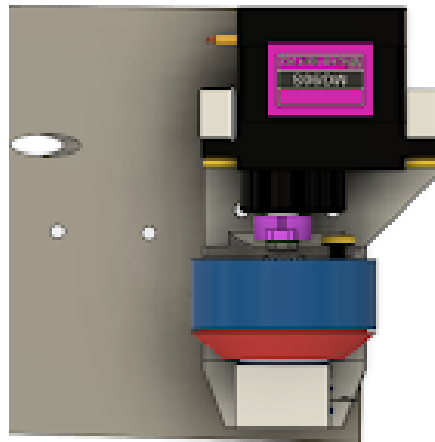


Figure 5.2: Servo on Right placed atop the String Guide Cover. Screw holes on both should line up.

4. Partially screw M2x6 screws into the screw holes in the Individual Servo Mount and the String Guide Cover so the end of the screw is barely poking out (Figure 5.3.b).

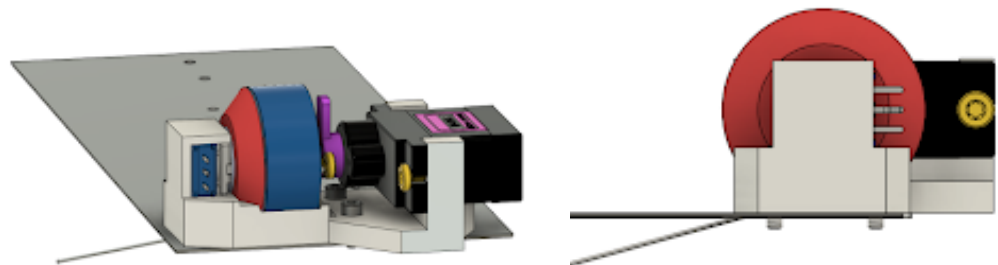


Figure 5.3.a (left): Servo on Right screwed into the String Guide Cover.

Figure 5.3.b (right): Side view of Figure 5.3.a. The screws should be barely out.

5. Take one Servo on Left and thread the loose end of the string through the closest next string hole.

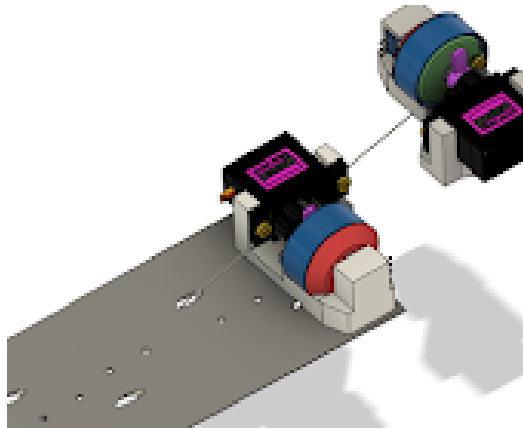


Figure 5.4: Servo on Left string threaded through the String Guide Cover next to the previously screwed Servo on Right.

6. Place the Servo on Left on the String Guide Cover, lining the corresponding screw holes on the String Guide Cover together.

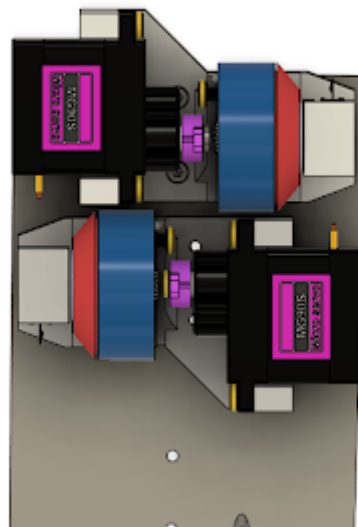


Figure 5.5: Servo on Left placed atop the String Guide Cover. Screw holes on both should line up.

7. Partially screw M2x6 screws into the screw holes in the Individual Servo Mount and the String Guide Cover so the end of the screw is barely poking out (Figure 5.6).

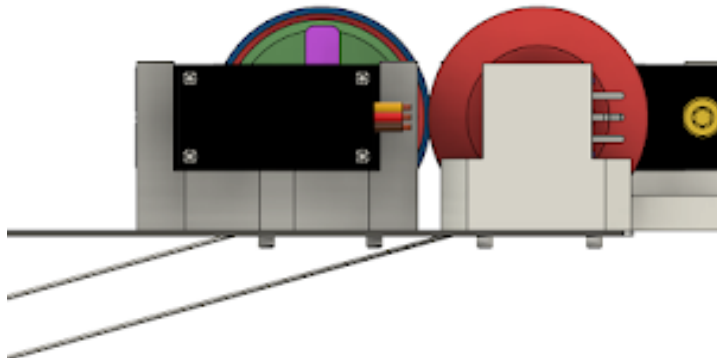


Figure 5.6: Servo on Right screwed into the String Guide Cover (to the left of Servo on Right). The screws should be barely out.

8. Repeat steps 2-7 for the next upward pair of string holes. There should be four servos on the String Guide Cover, all alternating (Figure 5.7.a).

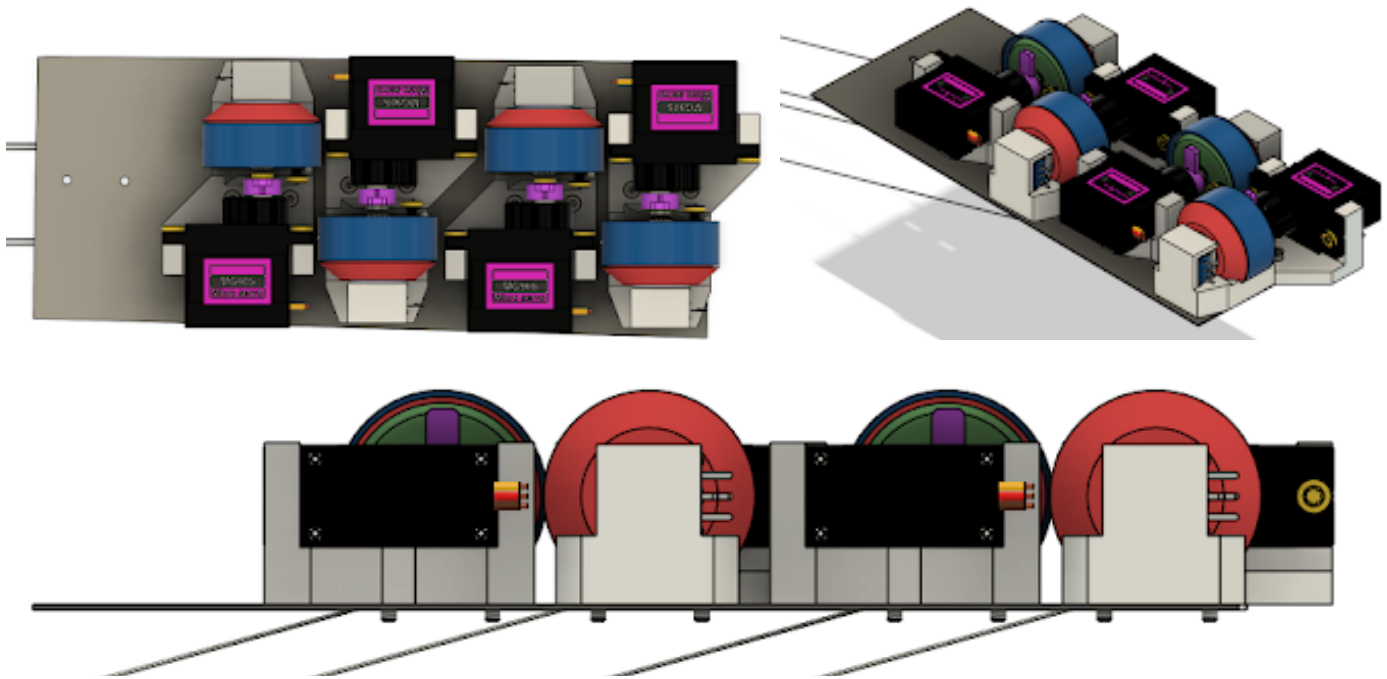


Figure 5.7.a (top left): All four servo components in the completed position, screwed into the String Guide Cover. The additional components: Servo on Right (first from left) and Servo on Left (second from left). All four alternate.

Figure 5.7.b (top right): Isometric view of the Figure 5.7.a String Guide Cover.

Figure 5.7.c (bottom): Side view of the Figure 5.7.a String Guide Cover.

9. For the last Servo on Left, simply align it with the final screw holes on the String Guide Cover so the string is going in the same direction as the rest

10. Partially screw M2x6 screws into the screw holes in the Individual Servo Mount and the String Guide Cover so the end of the screw is barely poking out.

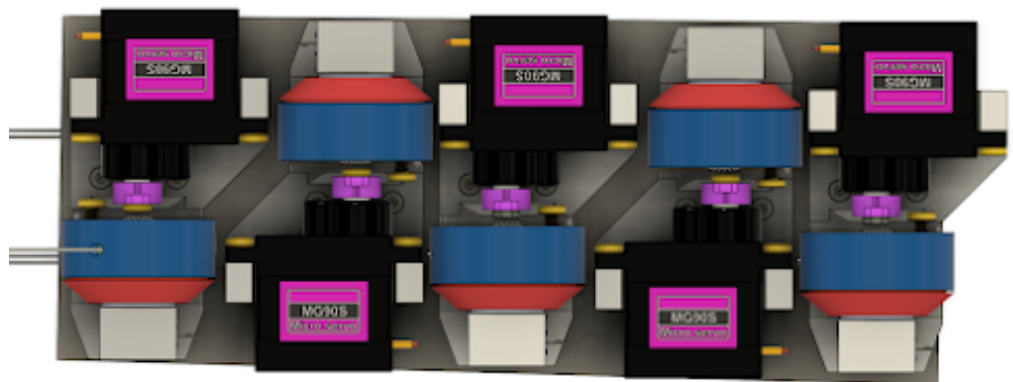


Figure 5.8: Top view of the completed String Guide Cover.

11. Place each string from the first 4 servo assemblies into their corresponding tunnel in the Servo Baseplate (Figure 5.9).

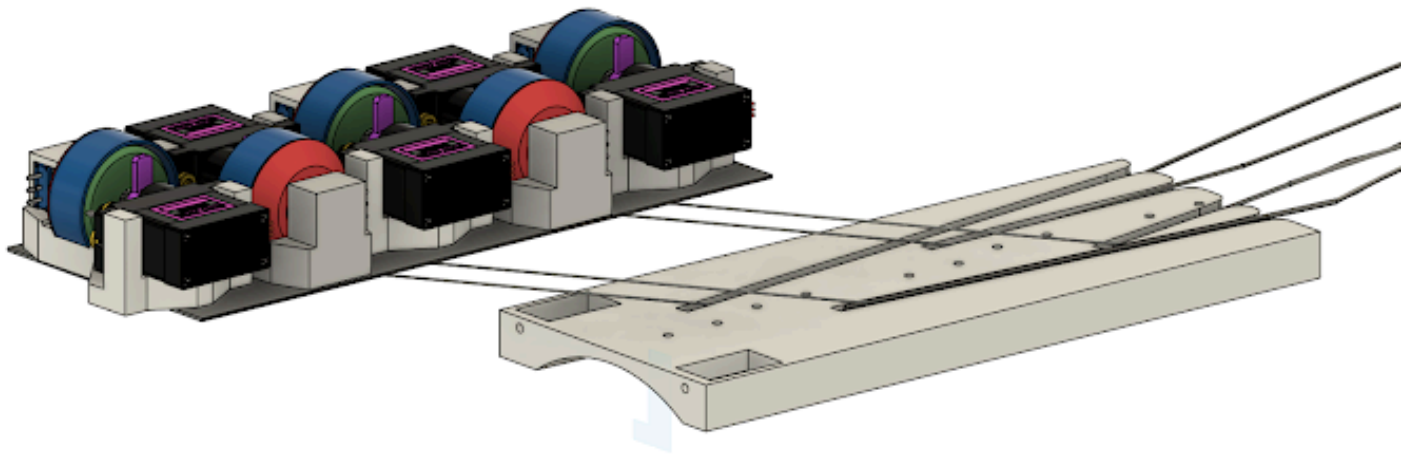


Figure 5.9: Each servo assembly string is threaded through the respective Servo Baseplate (Gray) tunnels.

12. Then line both atop one another so each partially outward screw will go into a respective hole on the Servo Baseplate, and screw all of them in fully.

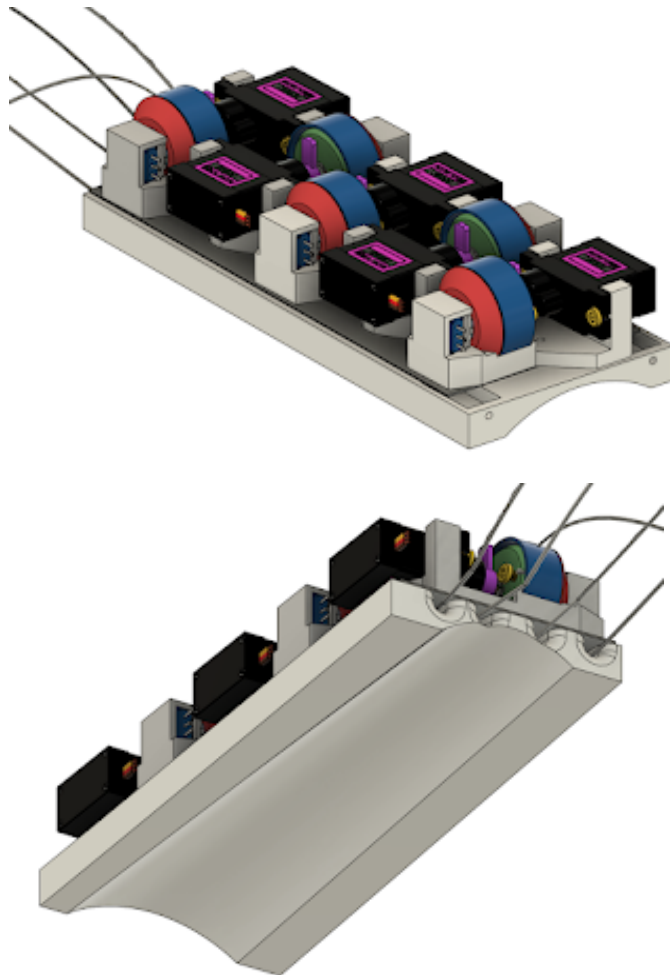


Figure 5.10.a (top): String Guide Cover atop the Servo Baseplate (Gray), completely screwed.

Figure 5.10.a (bottom): Bottom view of Figure 5.10.a

13. Plug two batteries with USB-C Female ports.



Figure 5.11: Battery connected to single USB-C female port.

14. Place both batteries into the Bottom Battery Mount and use a Sharpie to mark the positions that they line up with.

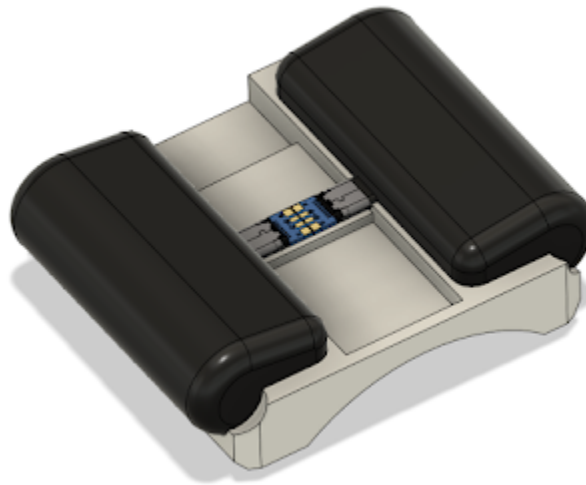


Figure 5.12: Both batteries placed within the Bottom Battery Mount (Gray). Each is connected with a USB-C female port.

15. Remove the batteries, keeping USB-C Female ports in place. Superglue both parts, with the 4 connectors facing up, into the spots you marked. Wait till dry.

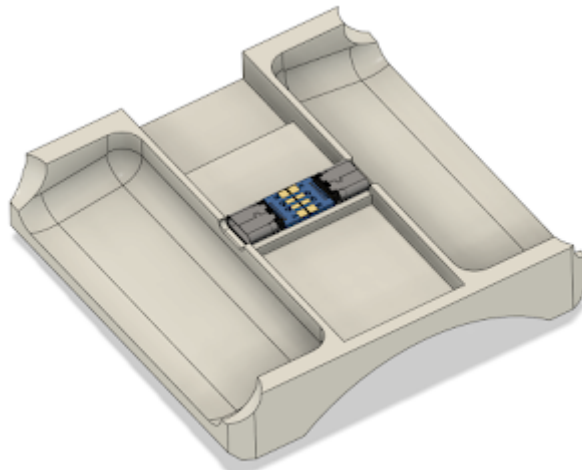


Figure 5.13: Both USB-C female ports are superglued onto the Bottom Battery Mount.

16. Place the Top Battery Mount on top of the Bottom Battery Mount (both sides should line up) and then superglue in place. USB-C female ports should be accessible through the outside mount hole (Figure 5.13.b). Wait till dry.

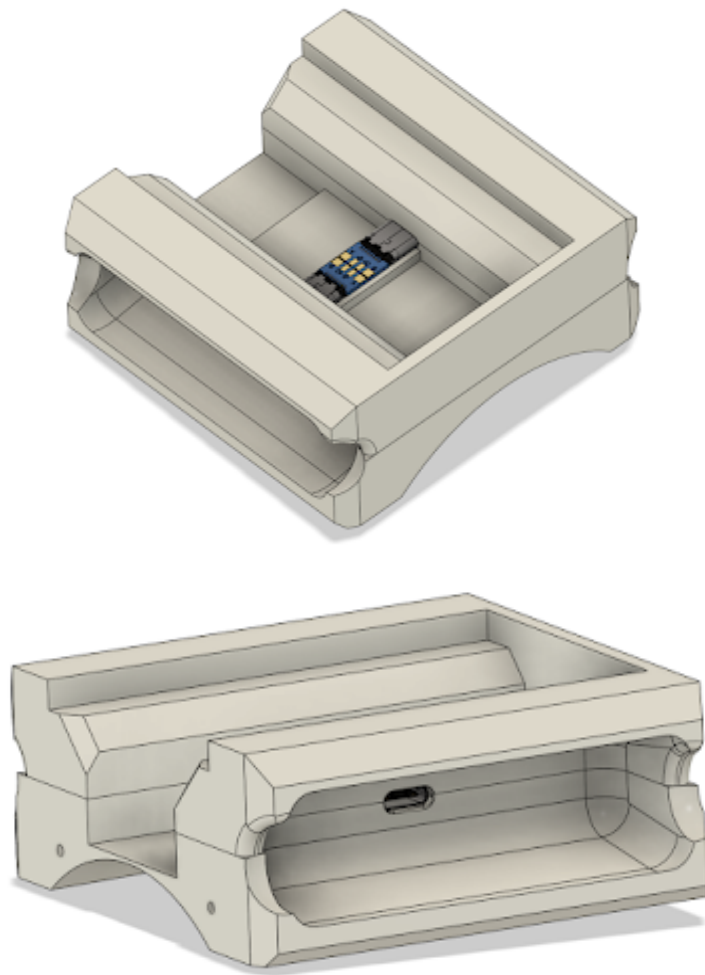


Figure 5.14.a (left): Complete Battery Mount part, with both USB-C female ports in place.

Figure 5.14.b (right): Alternate view of Figure 5.14.a. The female ports shall not be obstructed and be completely in line. Holes on the side of the Battery Mount are used to screw in place (Step 17).

17. With 2x M2.5x8 screws, connect the Servo Baseplate to the Bottom Battery Mount.

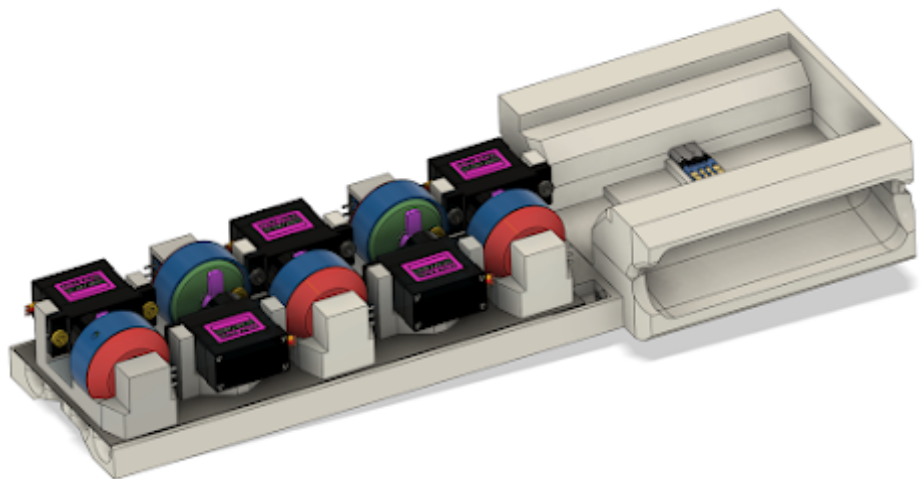


Figure 5.15: Complete part assembly of Servo Baseplate to Battery Mount.

18. With 4x M1.6x6 screws, thread each corner hole of the microcontroller, and attach it to the Microcontroller Holder. Set aside.

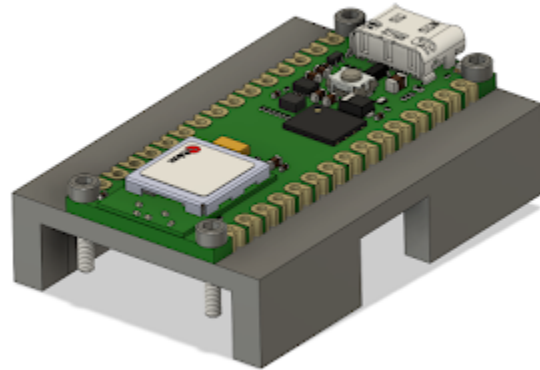
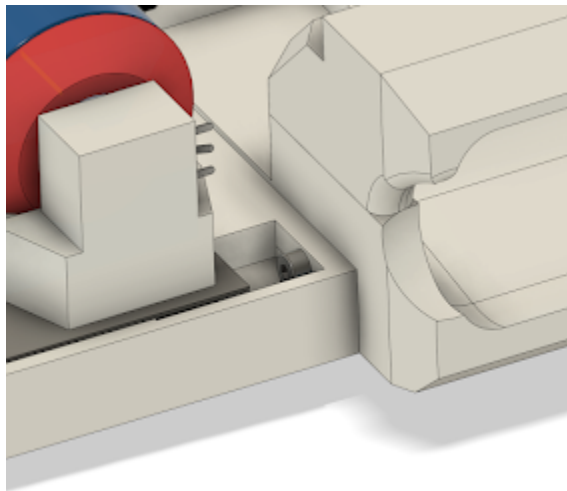
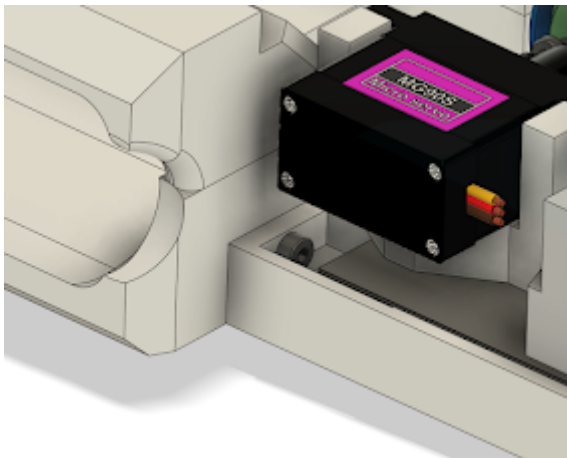


Figure 5.16: Complete part assembly of Microcontroller. This component will be used later.

19. With a sharpie, draw 2 marks on the Servo Base plate, 1 on each side. They should be ~1 inch from the front and 3/8th of an inch from the bottom.



20. Draw 2 more marks on the Servo Baseplate, 1 on each side. They should be ~ 1 inch from the Bottom Battery Mount, and 3/8th of an inch from the bottom (Figure 5.15).



21. Draw 2 more marks this time on the bottom battery mount, 1 on each side, ~ 1/8th of an inch under the middle of each battery slot.

Figure 5.17.a (top): The vicinity where one Step 20 mark would go.
Figure 5.17.b (bottom): The vicinity where the other Step 20 mark would go.

22. Superglue 6 metal loops (included with the velcro) to each spot you marked previously (Figure 5.18.a & 5.18.b). Each pair of metal loops on either side should completely line up.

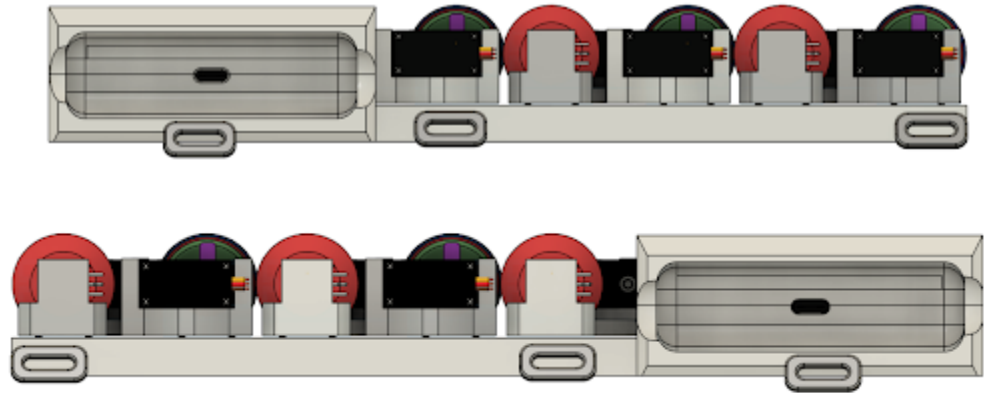


Figure 5.18.a (top): 3 metal loops attached to the right side of Servo Baseplate.
Figure 5.18.b (bottom): 3 metal loops attached to the left side of Servo Baseplate.

Final Assembly Appearance

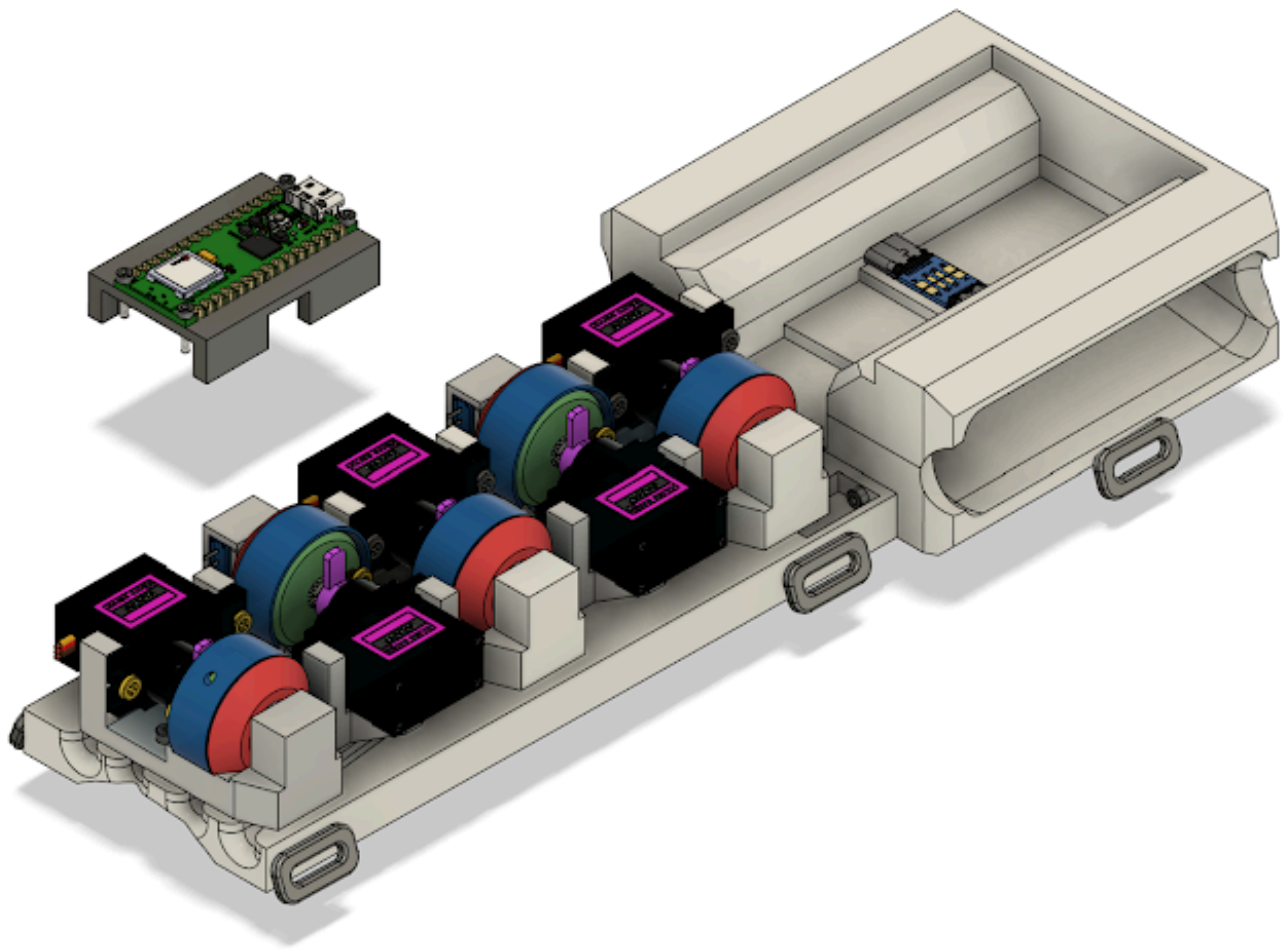


Figure 5.19: Complete part assembly of Electronics Holder.

The Electronics Holder components are now complete.

String Guide

1. Put the Latex Glove on.
2. With a sharpie, mark each joint in your fingers, including the knuckles, then take the glove off.
3. Superglue a String Guide to each one of the points you marked such that the hole through the String Guide is parallel to each finger.



Figure 6.1.a (top): Top view of Latex Glove with all String Guides attached.

Figure 6.1.b (bottom): Isometric view of 6.1.a.

4. Put the glove back on, and place the Quest Controller Holder on the glove so each tunnel end roughly lines up with a knuckle. Mark that position and take the Glove off.

5. Superglue the Quest Controller Holder onto the glove in the marked position.

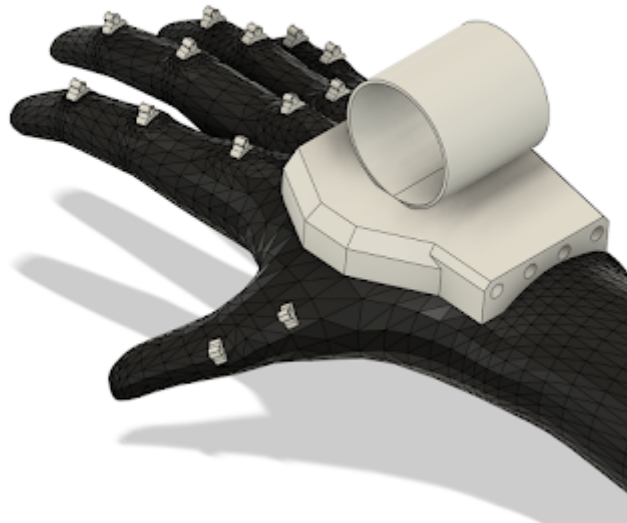


Figure 6.2.a (top): Top view of Latex Glove with all String Guides and Controller Holder attached.

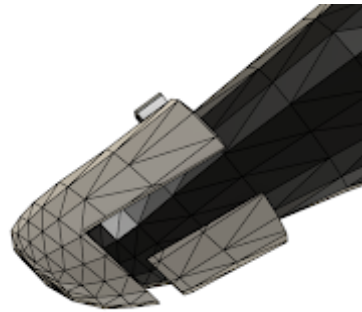
Figure 6.2.b (bottom): Isometric view of 6.2.a.

6. Take the top part of each finger (Thumb, Pointer, Middle, Ring, and Pinky) and super glue the String Holster onto the end of it. The String Holster hole must be parallel with its finger.



Figure 6.3: All Top Finger Parts, each with a String Holster.

7. Put the glove on, then place the Thumb Top so it rests on the tip of the thumb and the nail.



8. Place Thumb Bottom on the other side of the thumb.

Figure 6.3: Both Thumb Top and Bottom placed on finger.

9. Superglue one end of a 2 in. rubber band to the center of the top of Thumb Top.

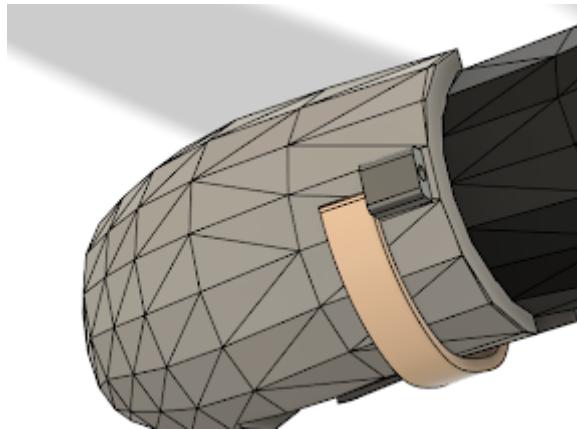


Figure 6.4: Both Thumb Top and Bottom placed on finger, with the rubber band attached.

10. Wrap the rubber band around the Thumb Top and Thumb Bottom, forming a loop. The rubber band should have a small amount of tension in it. If it is too long then cut a small portion off the end until it is the appropriate size.

11. Glue the other end of the rubber band next to the first end so it forms a single loop.

12. Find the middle of the rubber band and glue it to the middle of Thumb Bottom.

13. Take a 1-inch rubber band and glue one to the tip of the Thumb Top.

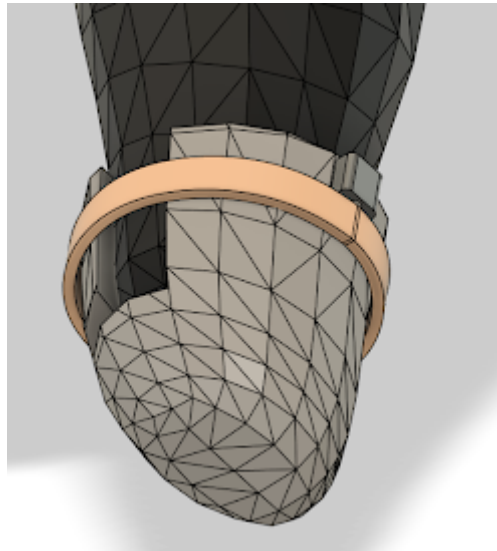


Figure 6.5: Both Thumb Top and Bottom placed on finger, with a looped rubber band attached.

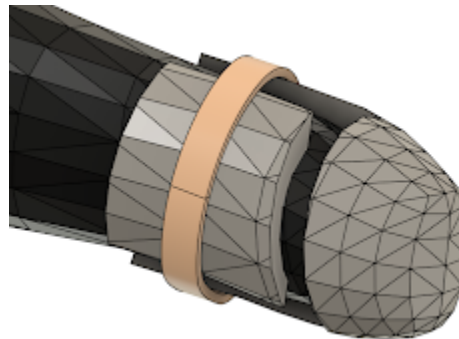


Figure 6.6: Both Thumb Top and Bottom placed on finger, rubber band attached to the bottom.

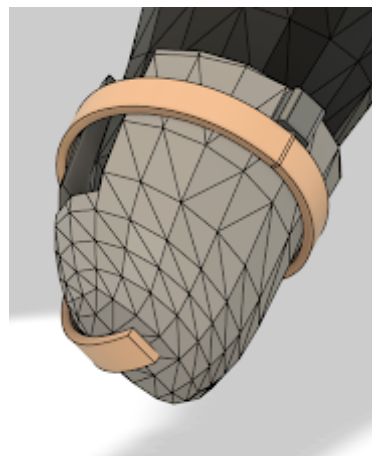


Figure 6.7: Additional rubber band attached to Thump Top.

14. Wrap it around the top of the thumb down to the bottom of the loop. If too long, shorten it so that it can touch the bottom of the rubber band loop while having some tension.

15. Once it is the appropriate length, superglue it in place.

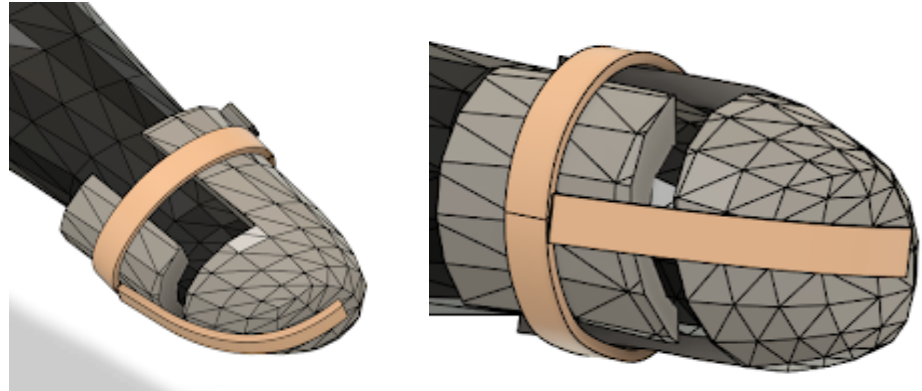


Figure 6.8.a (left): Thumb Mount.

Figure 6.8.b (right): Bottom view of Thumb Mount.

16. This assembly process is the same for each finger. Repeat the steps above but use the appropriate 3D-printed pieces for each finger.

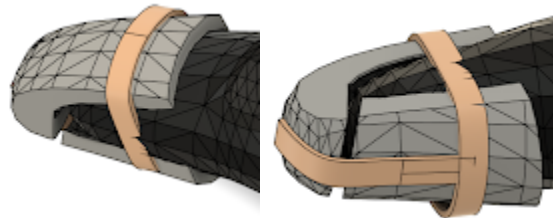


Figure 6.9.a: Pointer Finger Mount.

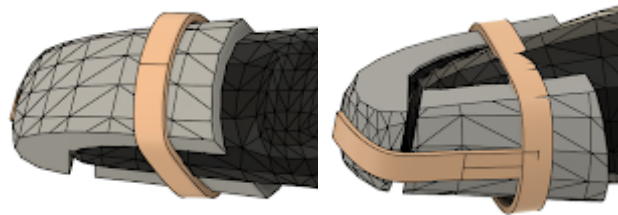


Figure 6.9.b: Middle Finger Mount.

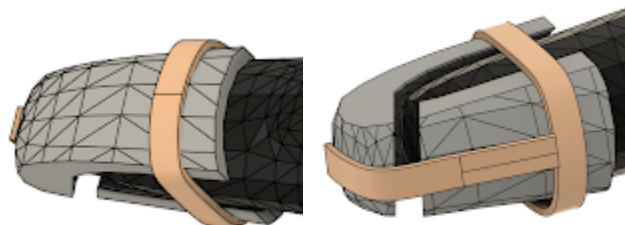


Figure 6.9.c: Ring Finger Mount.

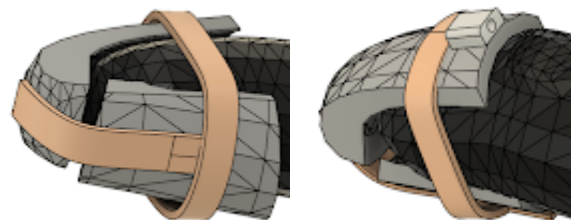


Figure 6.9.d: Pinky Finger Mount.

17. Superglue the top part of each Finger Mount to the Latex Glove directly below each string holster. Repeat for each finger.

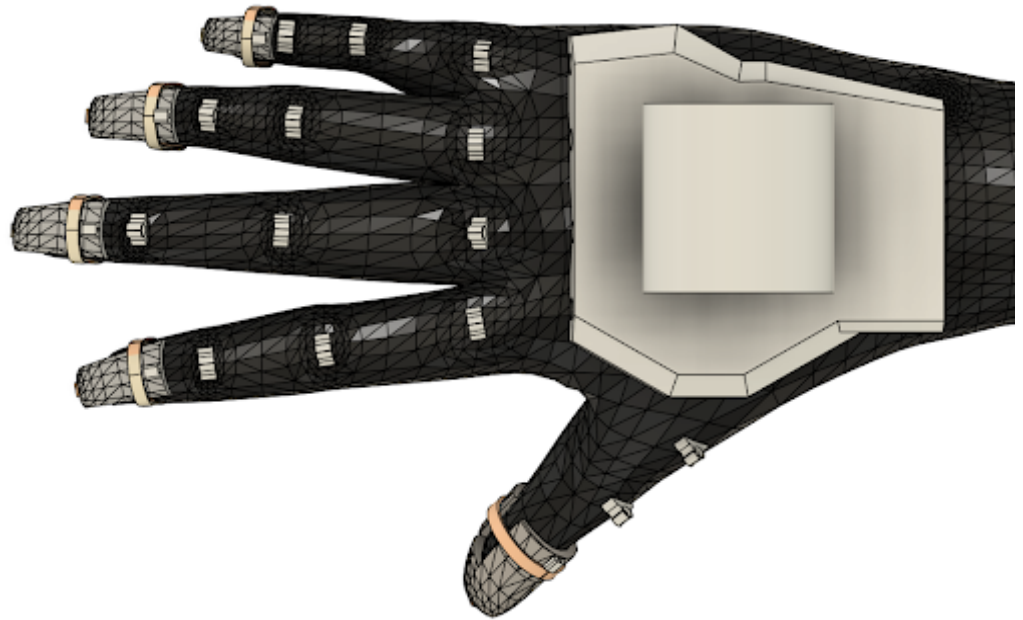


Figure 6.10: Top view of all Finger mounts attached to the Latex Glove.

18. Take two metal hoops (included with the velcro) and superglue them on both sides of the Oculus Quest Controller Holder, close to the back. Both should align.

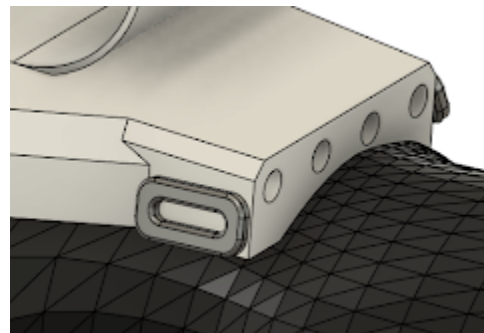
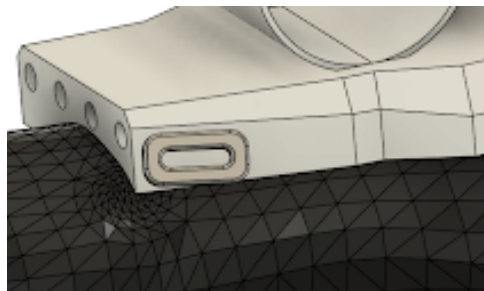


Figure 6.11.a (top): A metal loop attached to the right side of Quest Controller holder.

Figure 6.11.b (bottom): A metal loop attached to the right side of Quest Controller holder.

Final Assembly Appearance

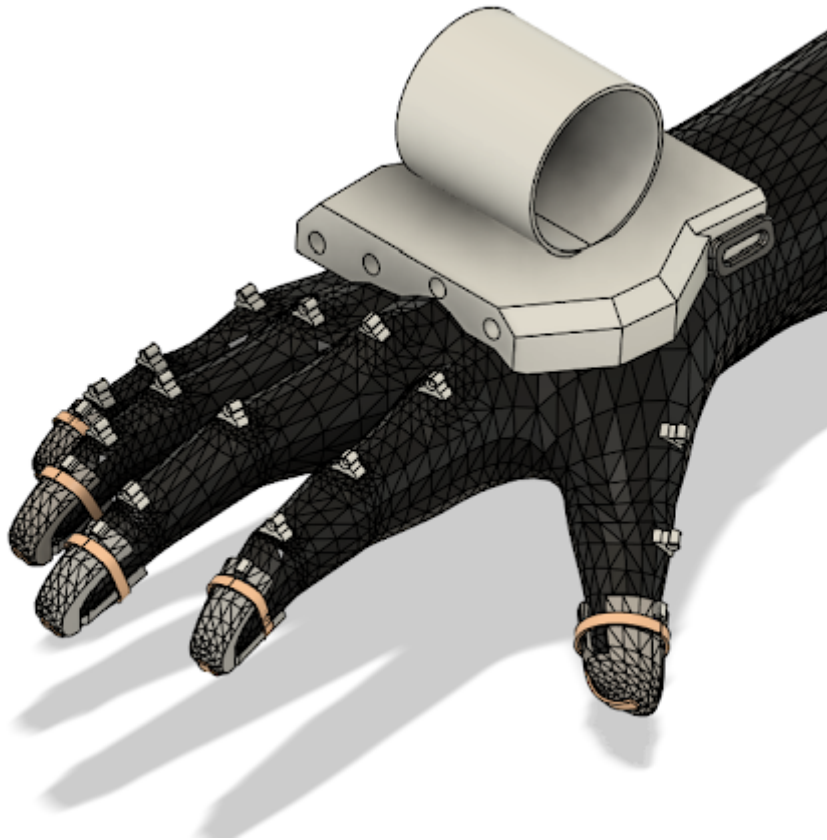


Figure 6.12: Top view of completed glove with string funnels and Oculus Quest Controller holder attached.

The String Guide component is now complete.

Wiring Guide

QUICK INFO

Reference Figure 7.0.a
& 7.0.b

Yellow wire is data.
Red wire is 5v in.
Black wire is ground.

The bottom prong is
the negative.
The middle (bent) prong
is data.
The top prong is the
positive terminal.

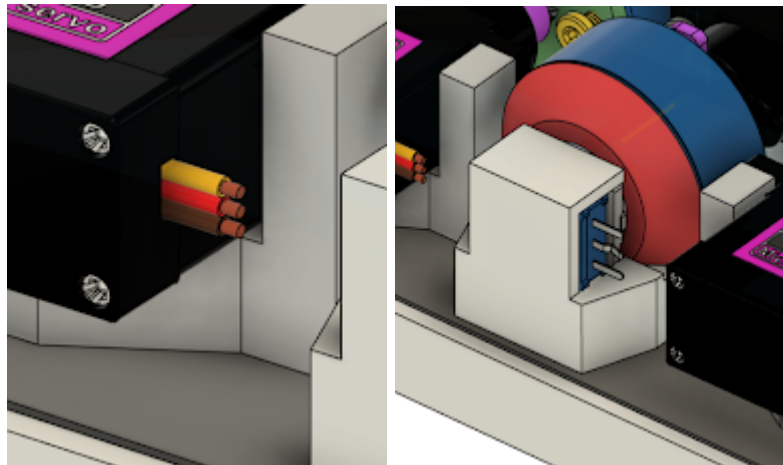


Figure 7.0.a (left): 3 wire outputs from Servo.

Figure 7.0.b (right): Three prongs from the potentiometer.

1. Move the end of
each wire connected to
the servos to the back
of the glove near the
battery holders.

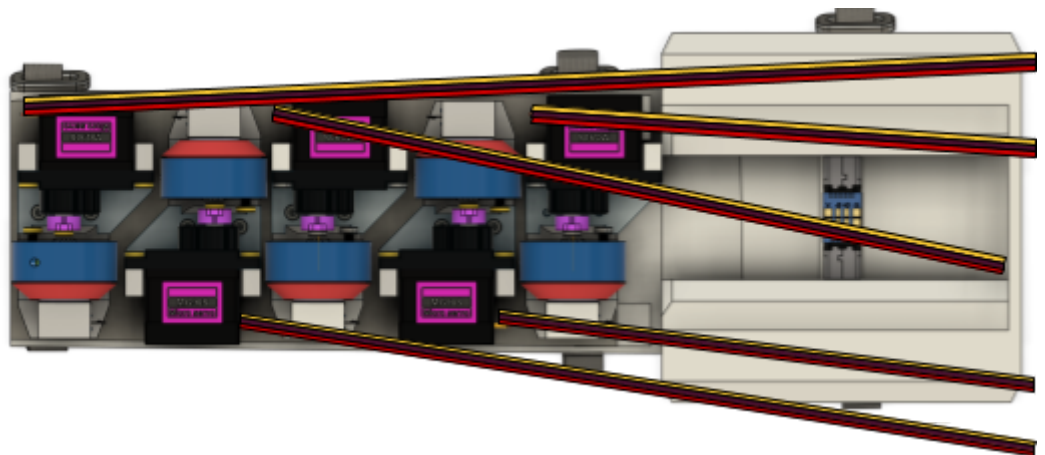


Figure 7.1: Cable Diagram with Servo Cables at back.

2. Cut the end off each
of the wires leaving as
much of the wire
connected, then
separate the 3 different
colors of cables.

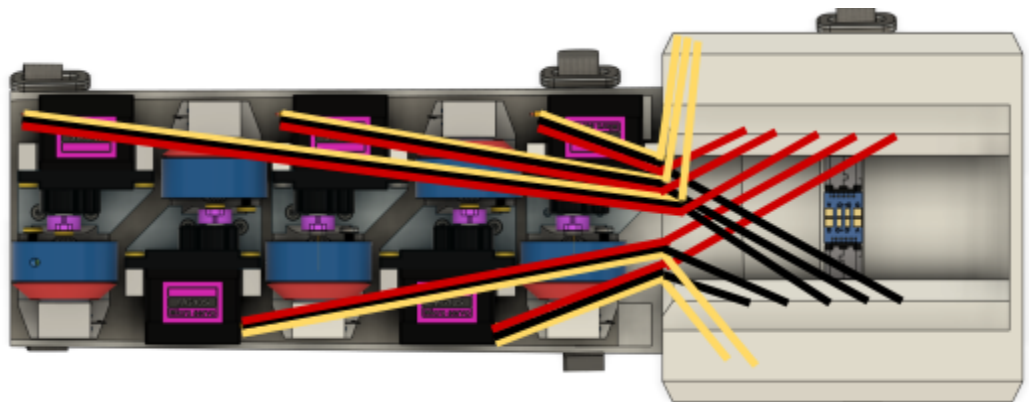


Figure 7.2: Cable Diagram with Servo Cables separated by type of wire.

3. With half an inch of red wire, connect the + terminals of the two USB-C Female ports.

4. With 1 inch of black wire, connect the - terminals of the two USB-C Female breakout boards.

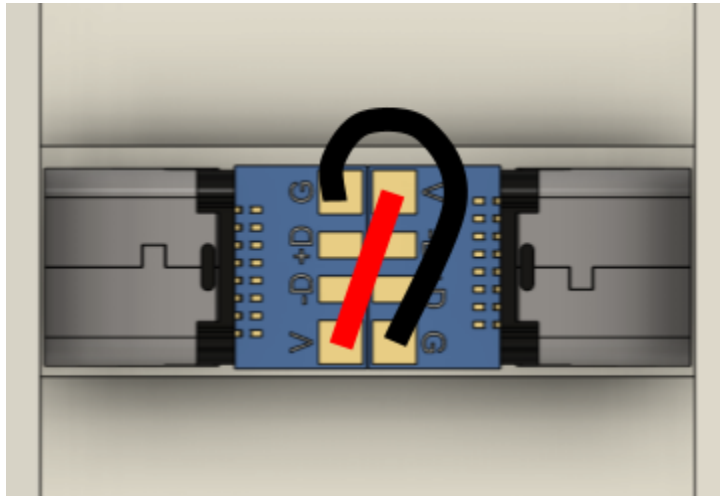


Figure 7.3: Wiring Diagram between two USB-C Female breakout boards.

Positive Servo

5. With 4 red wires, connect each of the positive wires from the servos (red wire) to the others with half an inch of red wire in between.

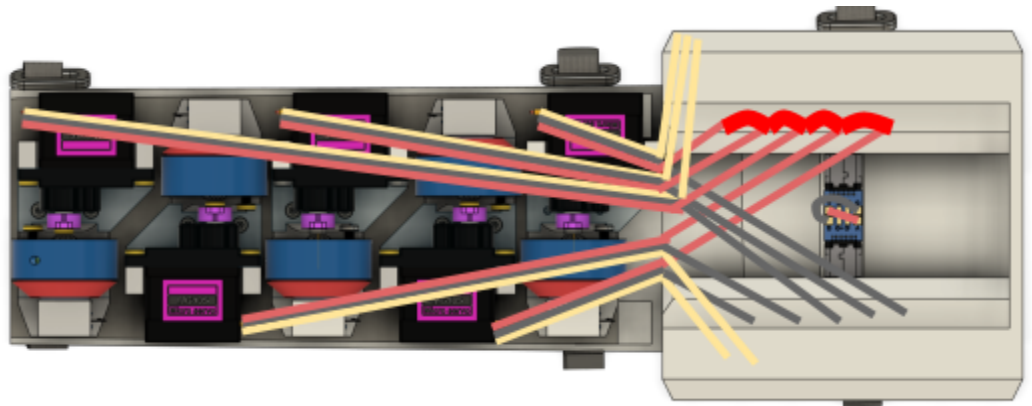


Figure 7.4: Wiring Diagram including connection between each 5v servo cable.
(Shown in high contrast red)

6. With a short red wire, connect this string of cables to the + pinout on the USB-C female breakout board.

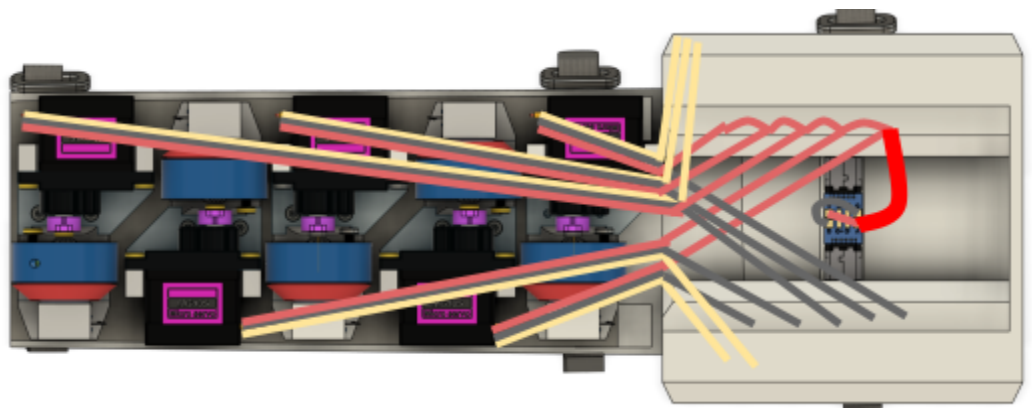


Figure 7.5: Wiring Diagram including connection between 5v servo wire and 5v out from USB-C Female Breakout Board. (Shown in high contrast red)

Positive Potentiometers

7. With 2 red wires, connect each of the positive terminals (top prong) of the potentiometers from the Servos on Right Assemblies to the others with ~ 4 in. of red wire in between.

8. With 1 red wire, connect each of the positive terminals (top prong) of the potentiometers from the Servos on Left Assemblies to the others with ~ 4 in. of red wire in between.

9. With a 4 in. red wire, connect the two positive strings of wires from the Servo on Left and Servo on Right assemblies together.

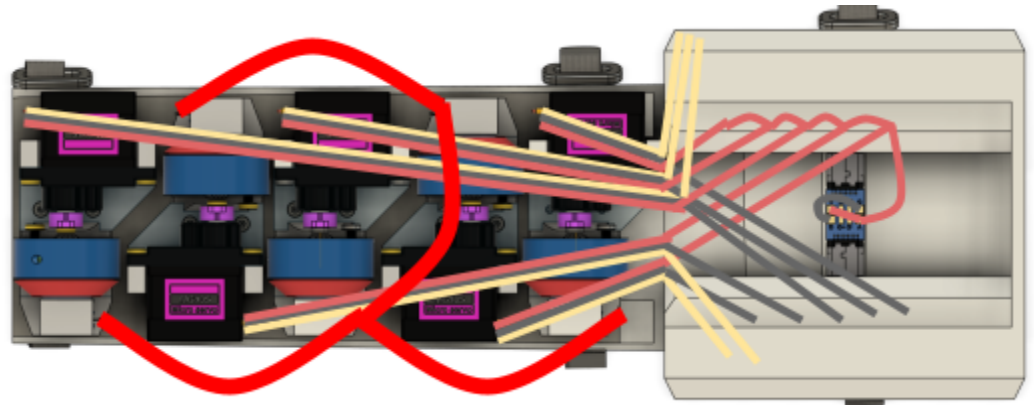


Figure 7.6: Wiring Diagram including connection between each positive terminal on the potentiometers. (Shown in high contrast red)

Negative Servos

10. With 4 black wires, connect each of the negative wires from the servos (black wire) to the others with half an inch of black wire in between.

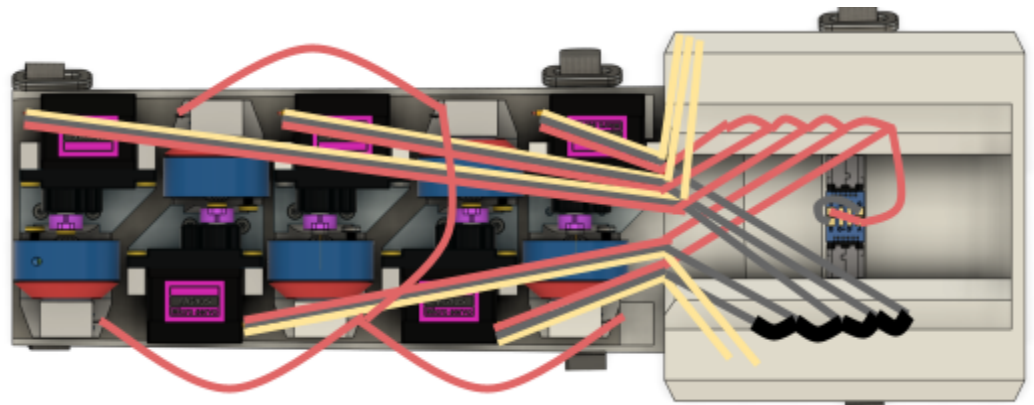


Figure 7.7: Wiring Diagram including connection between each ground servo cable. (Shown in Black)

11. With an appropriate length of black wire, connect the negative string of wires from the servo motors to the - terminal on the other USB-C female breakout board.

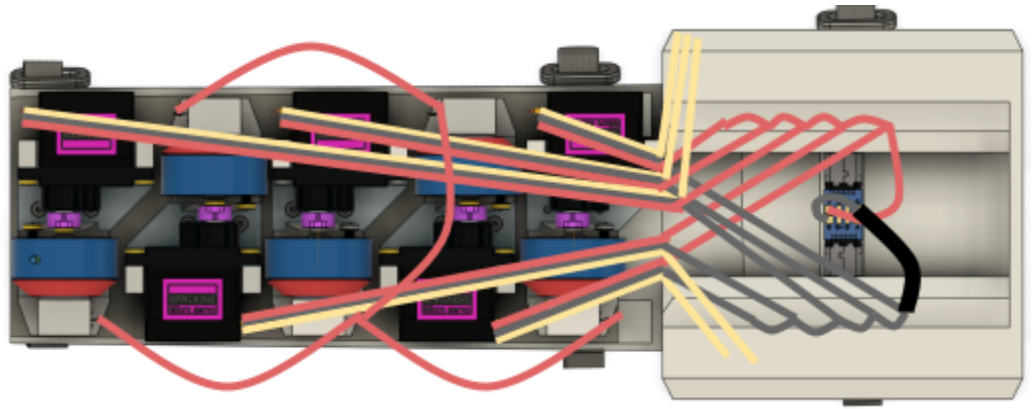


Figure 7.8: Wiring Diagram including connection between ground servo wire and - terminal from USB-C Female Breakout Board. (Shown in black)

Negative Potentiometers

12. With 2 Black wires, connect the negative terminals (bottom prong) of the potentiometers from Servos on Right Assemblies to the others with ~ 4 in. of black wire in between.

13. With 1 Black wire, connect the two negative terminals (bottom prong) of the potentiometers from Servos on Left Assemblies to the others with ~ 4 in. of black wire in between.

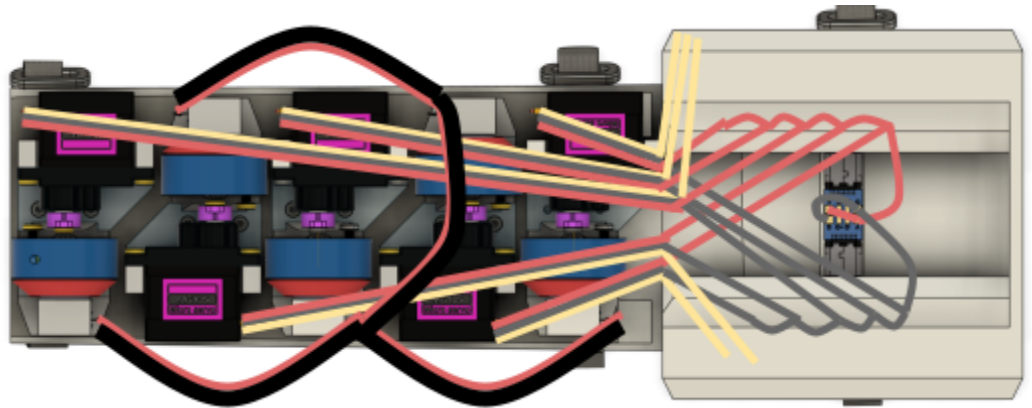


Figure 7.9: Wiring Diagram including connection between each negative terminal on the potentiometers. (Shown in black)

14. With an appropriate length of black wire, connect the negative string of wires from the potentiometers to the - terminal on the other USB-C female breakout board.

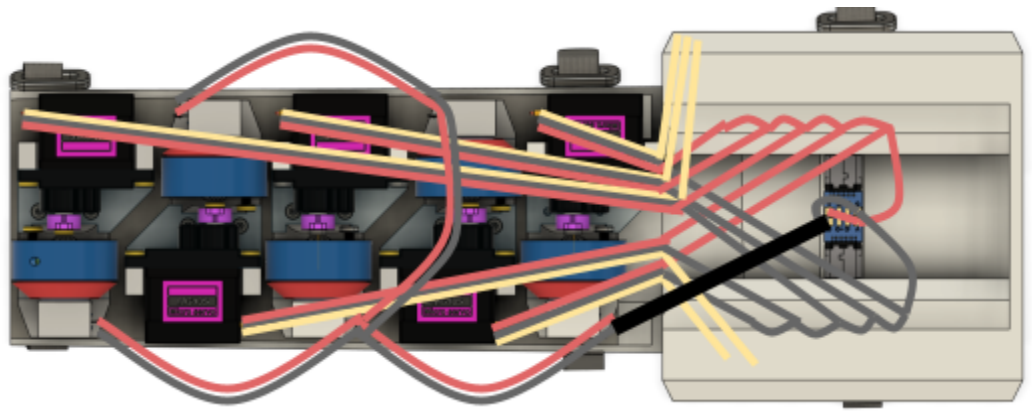


Figure 7.10: Wiring Diagram including connection between negative wire from potentiometers to - Terminal on USB-C Female Breakout Board. (Shown in black)

15. Solder one end of a 5-inch black wire to the negative terminal on either USB-C Female Breakout Board.

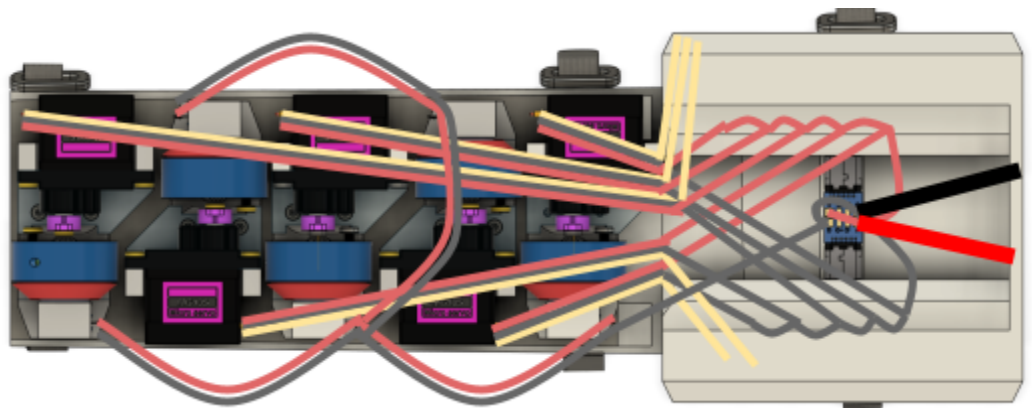


Figure 7.11: Wiring Diagram including 5v and GND cable from USB-C Female Breakout Boards. (GND in black, 5v in red)

17. Adjust the position of cables such that the microcontroller mount can go above the USB-C Female Breakout boards.

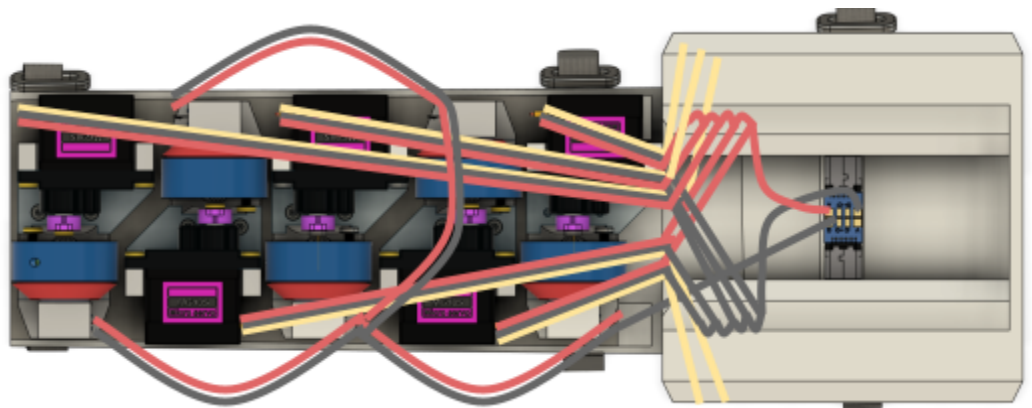


Figure 7.12: Wiring Diagram with adjusted cable positions.

18. Solder the loose end of the red cable from the USB-C Female port to the positive terminal on the USB-C Male Breakout board.

19. Solder the loose end of the black cable from the USB-C Female port to the positive terminal on the USB-C Male Breakout board.

20. Plug the USB-C Male breakout board into the USB-C Female port on the microcontroller.

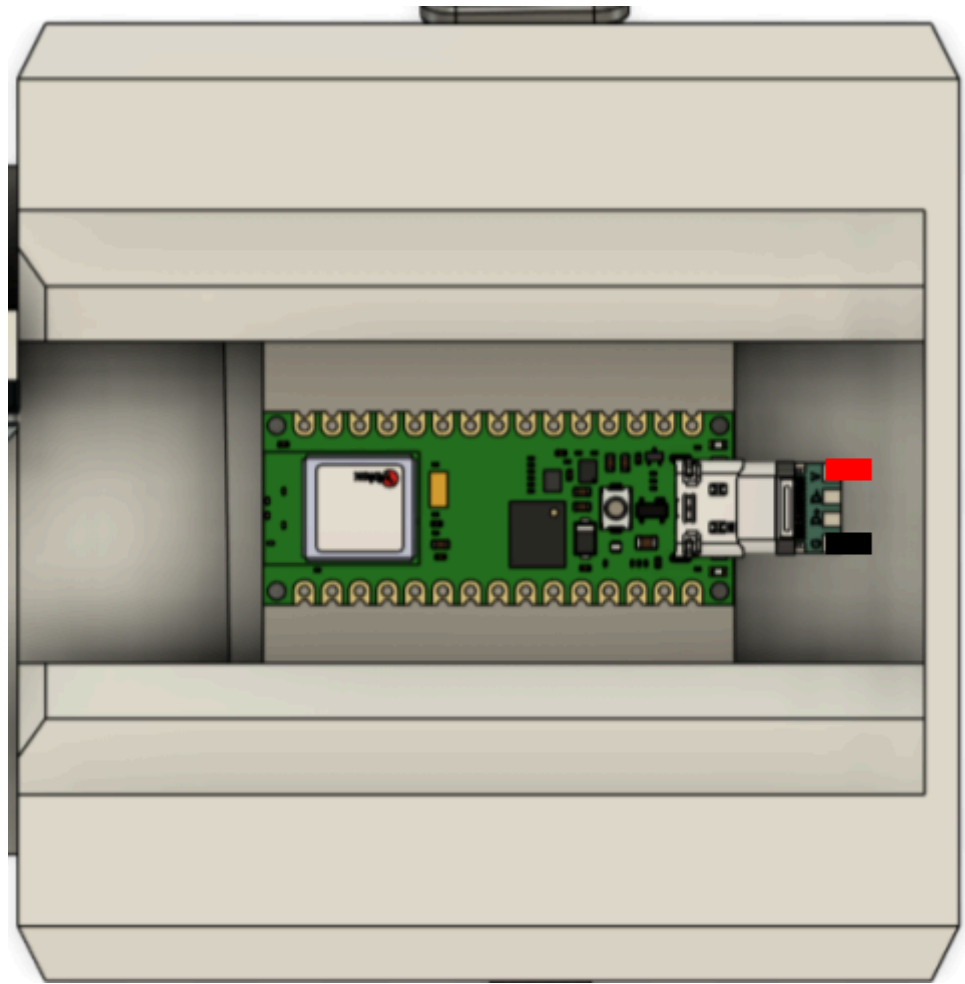


Figure 7.13: Microcontroller in correct position with USB-C Male Breakout Board Plugged in with 5v and GND connected.

21. With a 4 in. red wire, connect the positive wire from the potentiometers to the 3.3v pinout on the microcontroller. Use Green Wire when connecting the potentiometers to the microcontroller. Use Yellow Wire when connecting the servos to the microcontroller

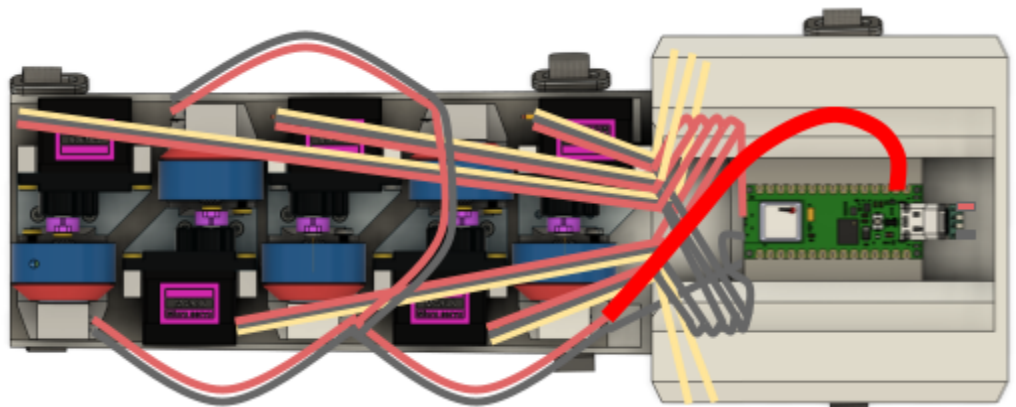


Figure 7.14: Wiring Diagram with 3.3v connected to positive cable from potentiometers. (Shown in high contrast red)

Pointer Data

22. Connect the middle prong of the potentiometer from the Servo on Right assembly closest to the batteries to the A4 pin on the microcontroller.

23. Connect the data cable from the servo from the Servo on Right assembly closest to the batteries to the D5 pin on the microcontroller.

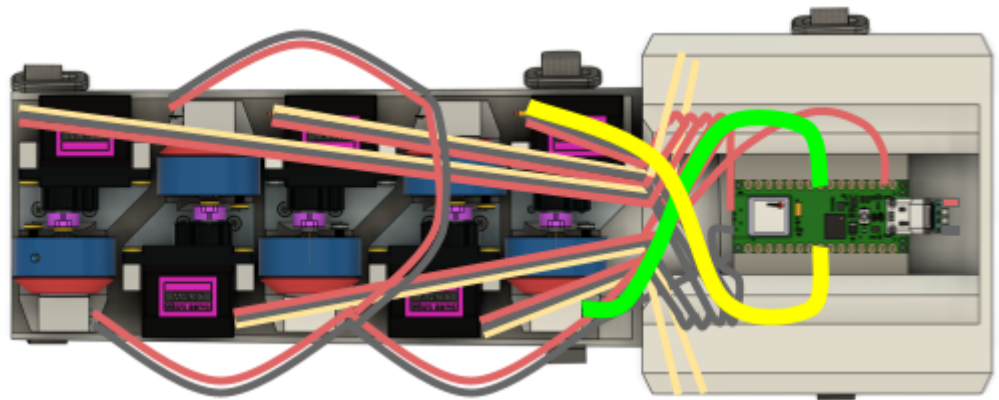


Figure 7.15: Wiring Diagram with one servo and potentiometer connected. (Servo data in yellow, Potentiometer data in green)

Pinky Data

24. Connect the middle prong of the potentiometer from the Servo on Left assembly closest to the batteries to the A1 pin on the microcontroller.

25. Connect the data cable from the servo from the Servo on Left assembly closest to the batteries to the D2 pin on the microcontroller.

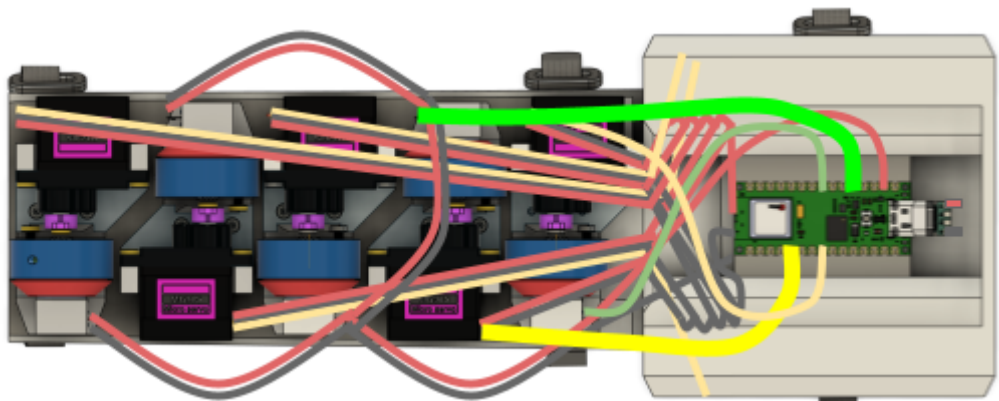


Figure 7.16: Wiring Diagram with two servos and potentiometers connected. (Servo data in yellow, Potentiometer data in green)

Middle Data

26. Connect the middle prong of the potentiometer from the middle Servo on Right assembly to the A3 pin on the microcontroller.

27. Connect the data cable from the servo from the middle Servo on Right assembly to the D4 pin on the microcontroller.

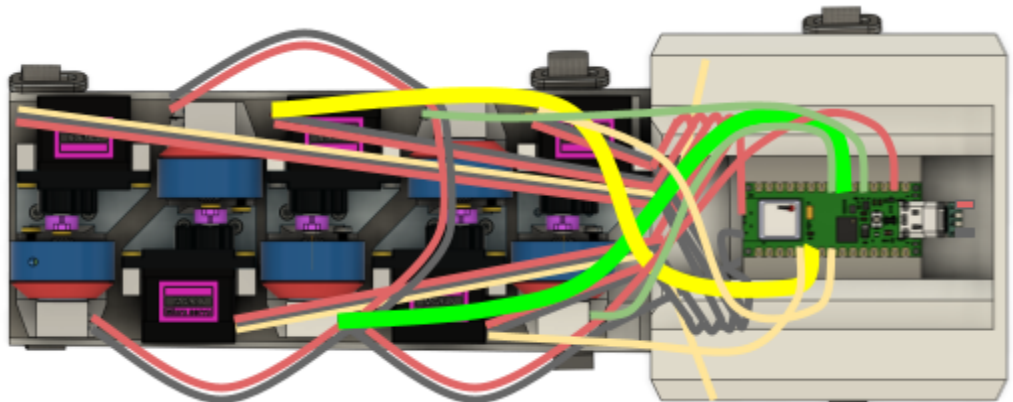


Figure 7.17: Wiring Diagram with three servos and potentiometers connected. (Servo data in yellow, Potentiometer data in green)

Ring Data

28. Connect the middle prong of the potentiometer from the next Servo on Left assembly to the A2 pin on the microcontroller.

29. Connect the data cable from the servo from the next Servo on Left assembly to the D3 pin on the microcontroller.

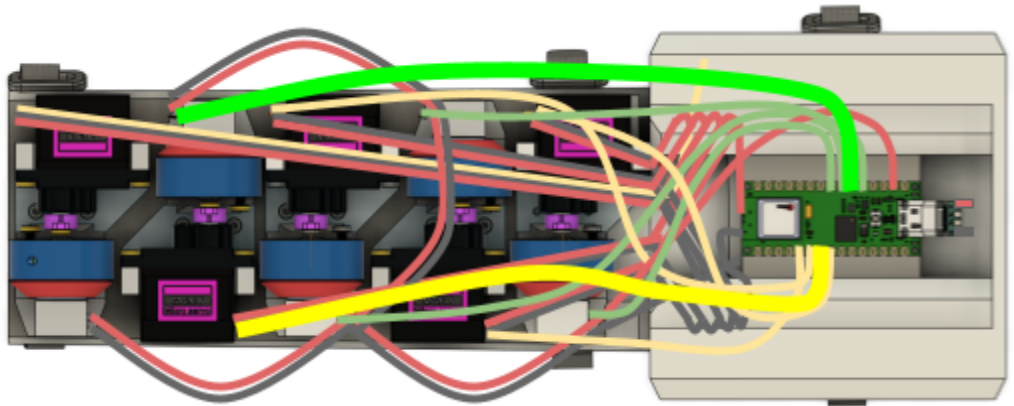


Figure 7.18: Wiring Diagram with four servos and potentiometers connected. (Servo data in yellow, Potentiometer data in green)

Thumb Data

30. Connect the middle prong of the potentiometer from the last Servo on Right assembly closest to the batteries to the A5 pin on the microcontroller.

31. Connect the data cable from the servo from the last Servo on Right assembly to the D6 pin on the microcontroller.

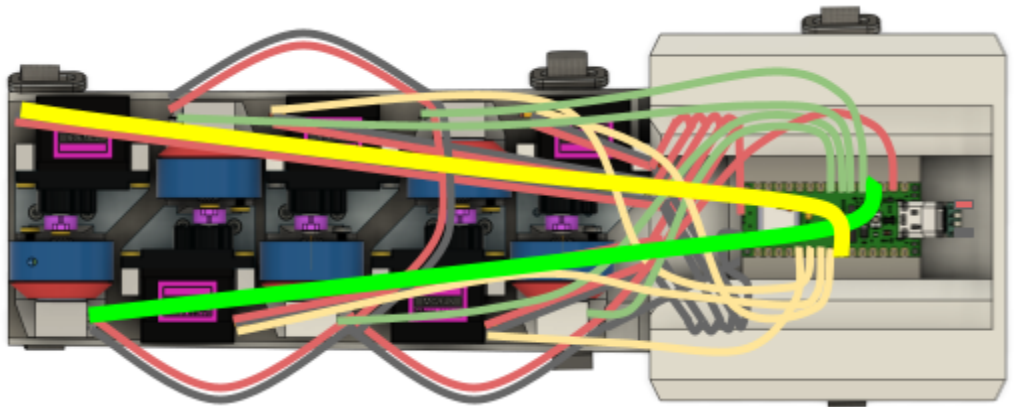


Figure 7.19: Wiring Diagram with five servos and potentiometers connected. (Servo data in yellow, Potentiometer data in green)

Final Steps

Cables will not be shown in these instructions, but wiring should be completed before these steps.

1. Put the glove on with the Electronics Holder in a comfortable position on the arm, facing upwards with the string going out of the mount in the correct direction.

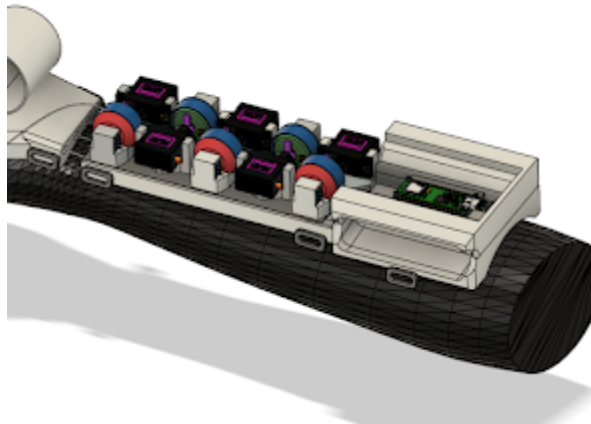


Figure 8.1: Electronics Holder on Latex Glove.

2. Cut 3x ~12 in. of velcro off of the reel. And thread it through the velcro loops on the left side of the glove.
3. Connect the velcro to itself so it is connected to the glove, then superglue it closed.

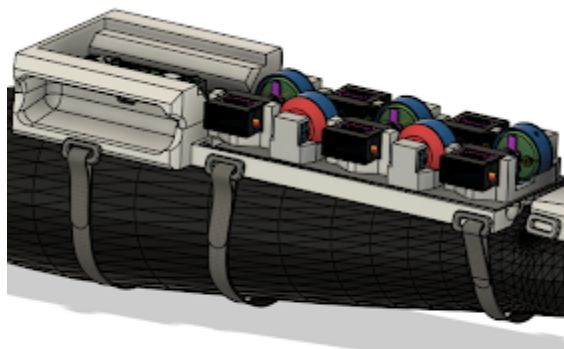


Figure 8.2: Electronics Holder with Added Velcro onto Latex Glove.

4. Slide each velcro piece through the corresponding loop on the other side and pull it tight so the velcro attaches the electronics mount to the arm.

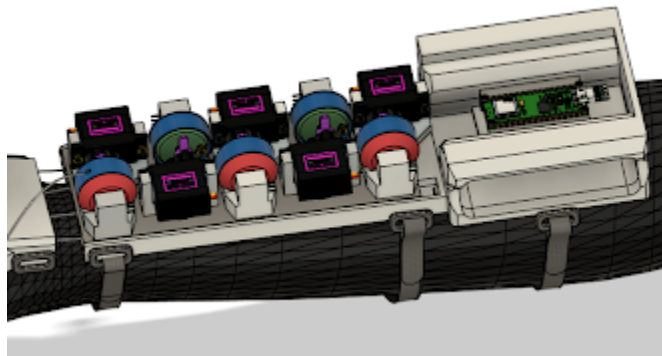


Figure 8.3: Electronics Holder with Fastened Velcro onto Latex Glove.

5. Follow steps 2-5 but with a single piece of 10-inch velcro, with the loops on the Oculus Quest Controller Holder.

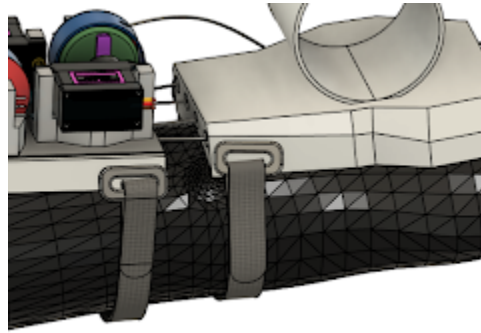


Figure 8.4: Closer view of Electronics Holder with Added Velcro onto Quest Controller Holder.

6. For the 4 strings in a string tunnel, put the string through the closest hole in the Oculus Quest Controller Holder.

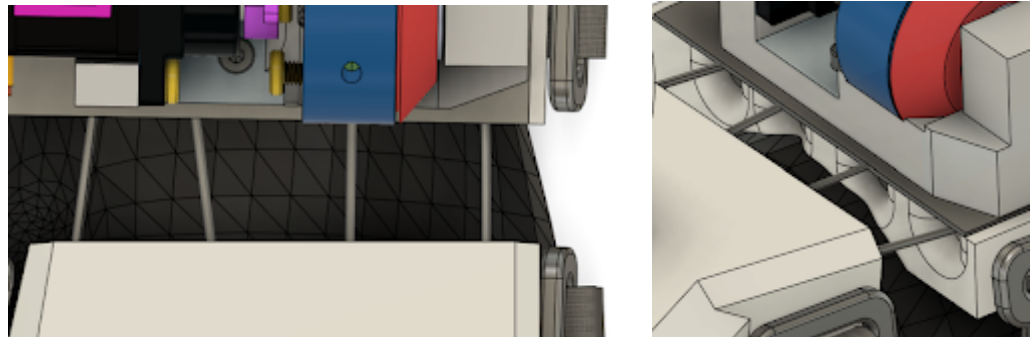


Figure 8.5.a (left): Four wires threaded through the holder.

Figure 8.5.b (right): Isometric view of 8.5.a.

7. For the one string not in a string tunnel, put the string through the hole in the String Top Cover, then place the string top cover onto the String Base plate.

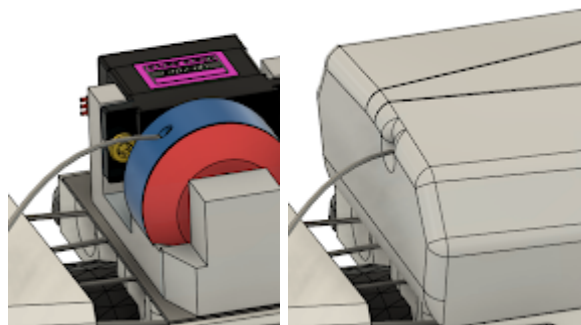


Figure 8.6.a (left): String threaded through hole in String Top Cover with cover not visible.

Figure 8.6.b (right): String threaded through hole in String Top Cover with cover visible.

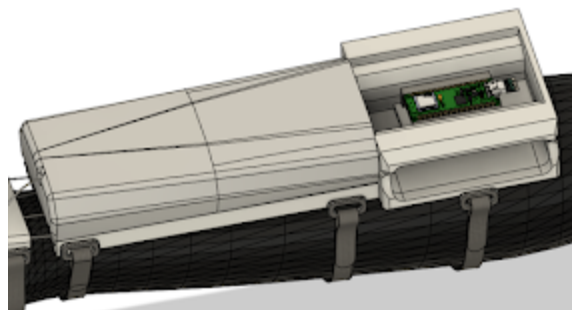


Figure 8.6.c: View of full String Base plate.

8. Move the rest of this string through the finger guides located on the thumb then through the string holster on Thumb Top. Then Tight a knot on the other side of Thumb Top so the string is connected to Thumb Top.

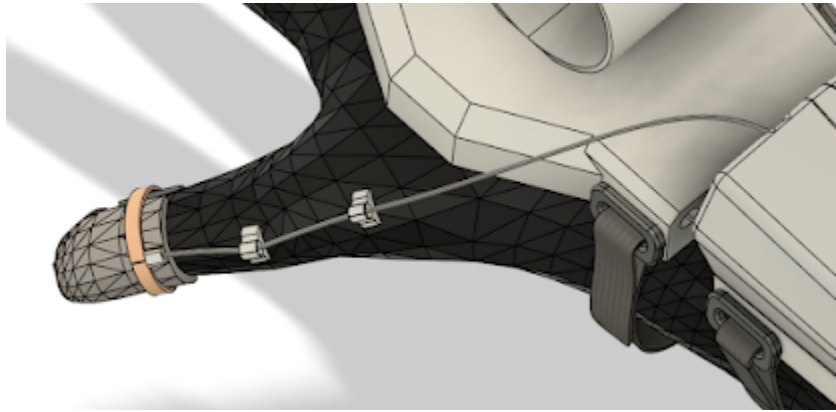


Figure 8.7.a: View of threaded thumb string.

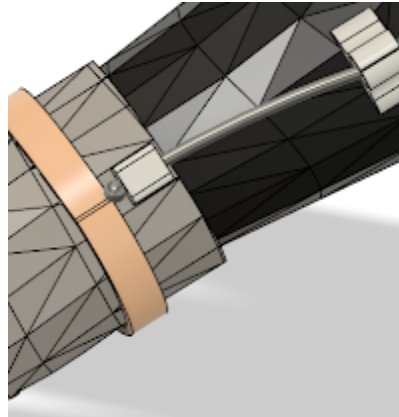


Figure 8.7.b: Close view of string connected into Thumb Top holster (Light Gray).

9. Take each of the 4 strings coming out of the Oculus Quest Controller Holder, and thread them through the string guides on the corresponding fingers.

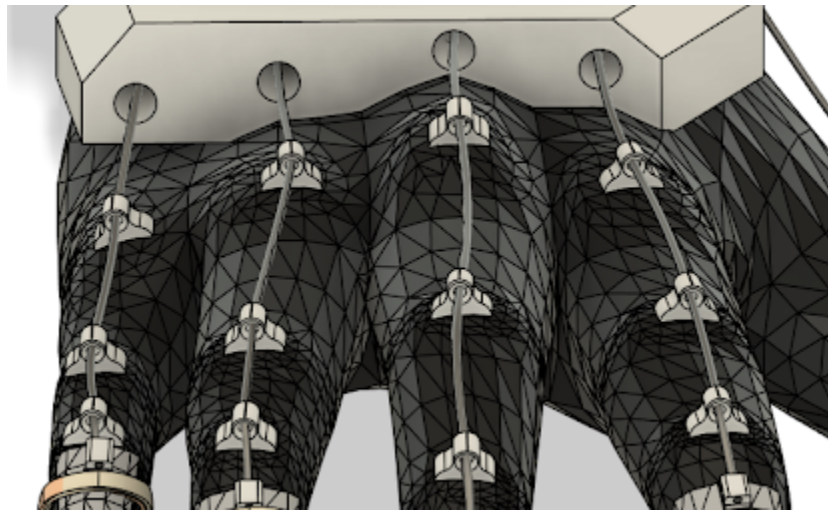


Figure 8.8: Front view of string threaded through each correct hole.

10. Take those 4 strings and thread the strings through the corresponding String Holsters, then tie a knot directly on the other side of the Holsters.

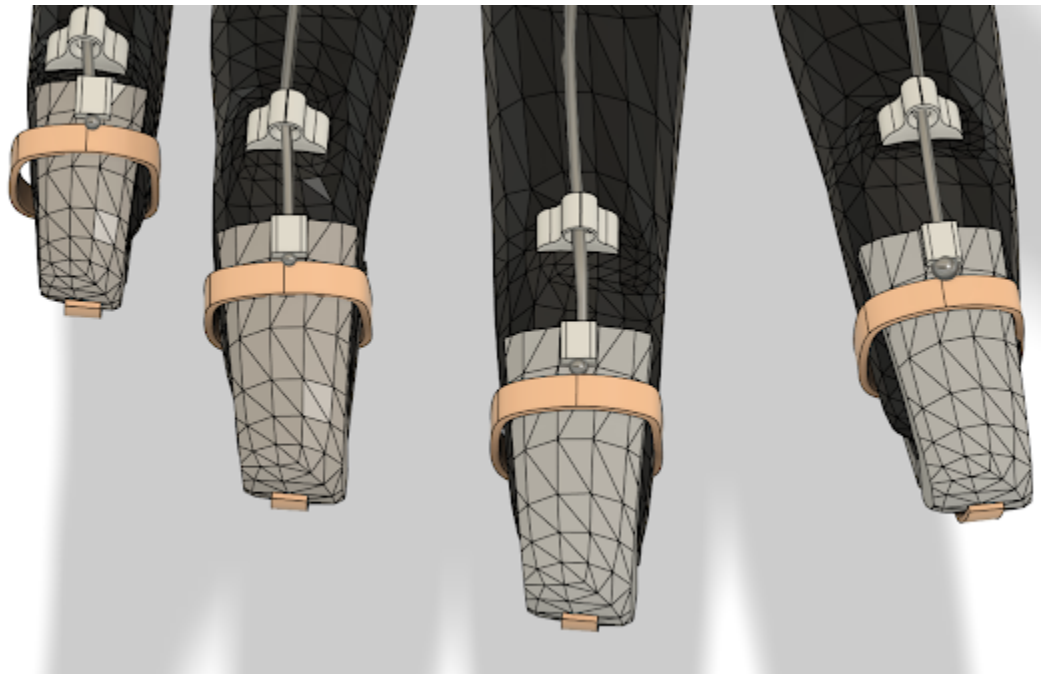


Figure 8.9: Front view of strings attached to corresponding holsters.

11. Place the Servo Top cover into its position above the Top Battery Mount.

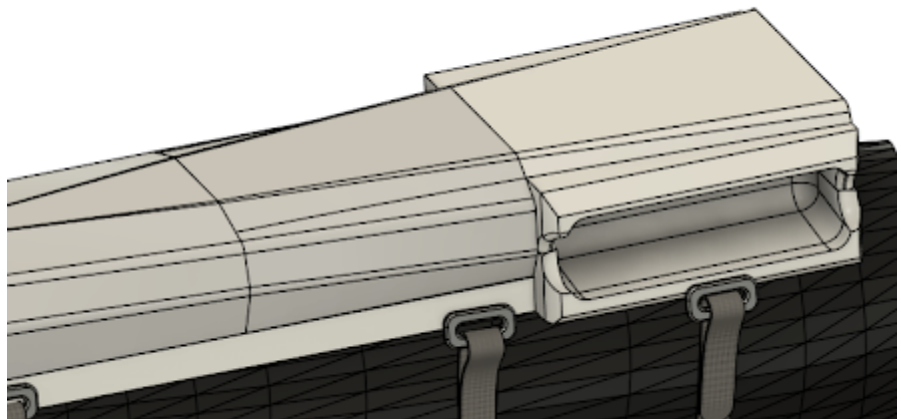


Figure 8.10: Servo Top Cover attached onto the Top Battery Mount.

12. Put the Oculus Quest Controller into the cylindrical shell on top of the Oculus Quest Controller Holder.

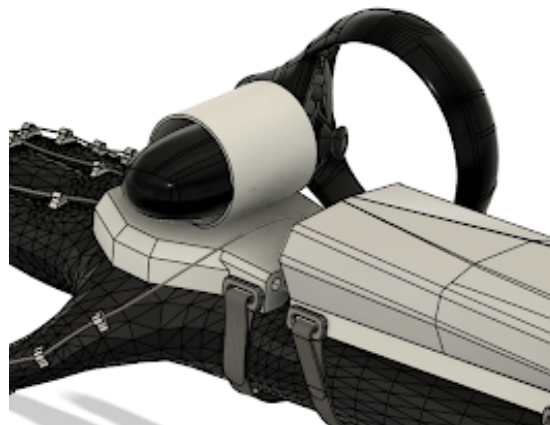


Figure 8.11: Oculus Question Controller put into place.

13. Lastly, plug the batteries in.

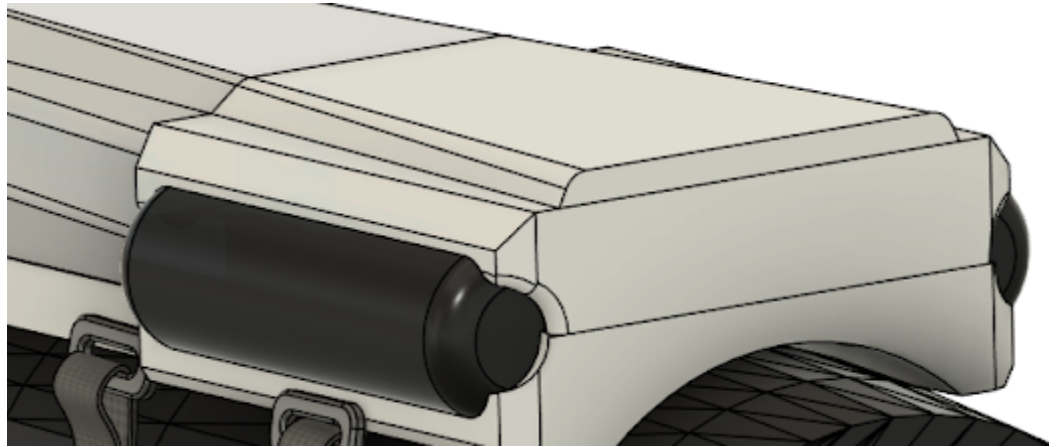
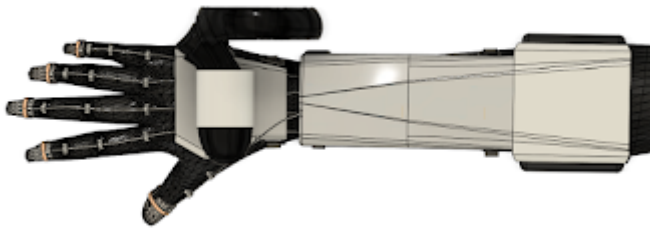


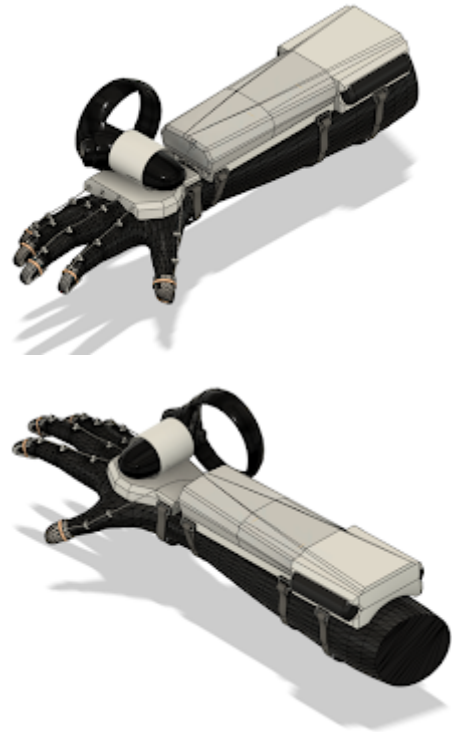
Figure 8.12: Battery Mount with batteries attached.

IV. Complete Assembly

Top



Isometric



Front



Side

